

PHNIX

Swimming Pool Heat Pump

Technical Manual



PHNIX reserves the right to discontinue, or change at any time, specifications or designs without notices and without incurring obligations.

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The cleanness of inner pipe (mg/m ²)	错误!未定义书签。
Hydrostatic testing (MPA).....	错误!未定义书签。
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Section 1 the Summary of Product

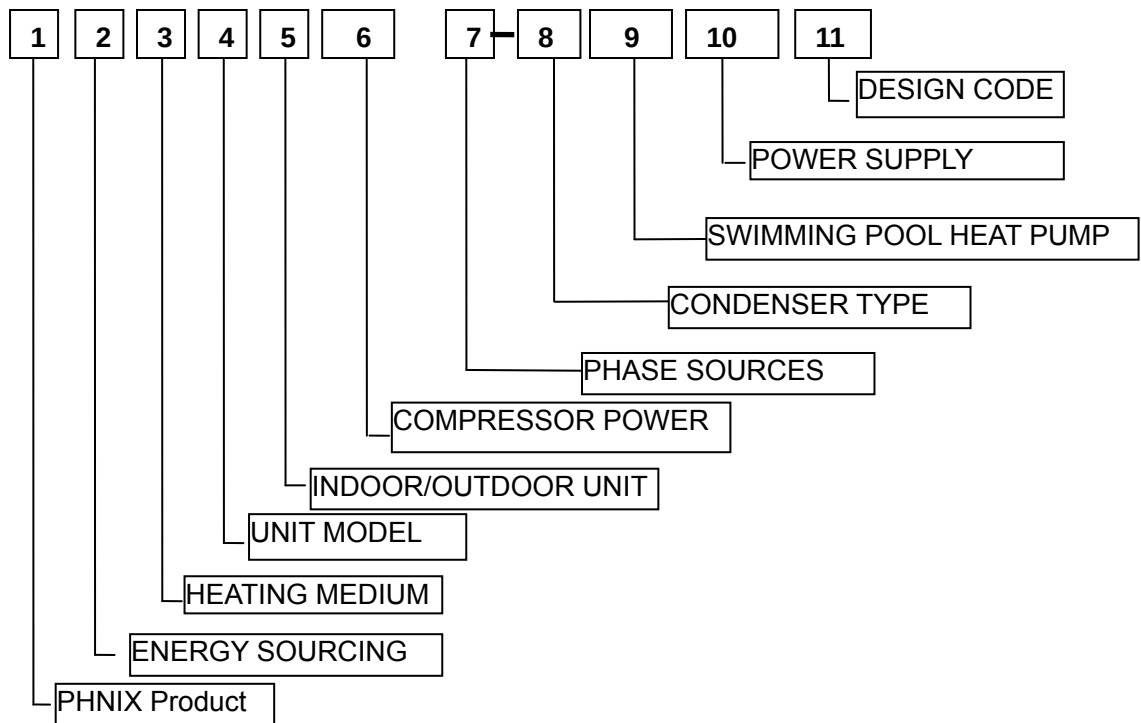
1. Preface

- ◆ Thank you for your trust and reliance on PHNIX and choosing and purchasing PHNIX Swimming pool heat pumps.
- ◆ The manual includes necessary information concerning correct installation, modification, running and maintenance of units. Before running or mending machines, please carefully read this manual.
- ◆ When you are installing swimming pool heat pump, connecting water circulation and wiring electric or electronic devices, please strictly operate on this manual. After finishing installation and checking, then power on machines.
- ◆ PHNIX reserves rights to change specification and design of units which leads to content change of this manual.

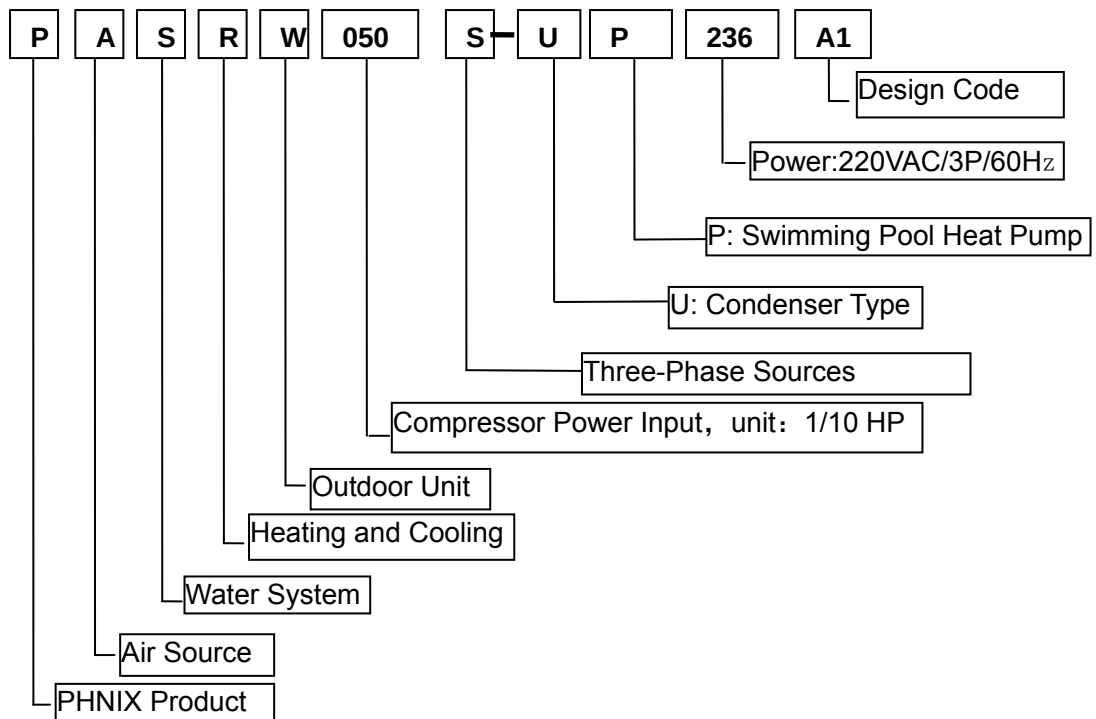
2. Product presentation

PHNIX swimming pool heat pump can be used for heating and cooling for swimming pool, it has operate easy, work safety, high efficiency, environment friendly, long life work characteristic and so on.

3. Nomenclature



For example:

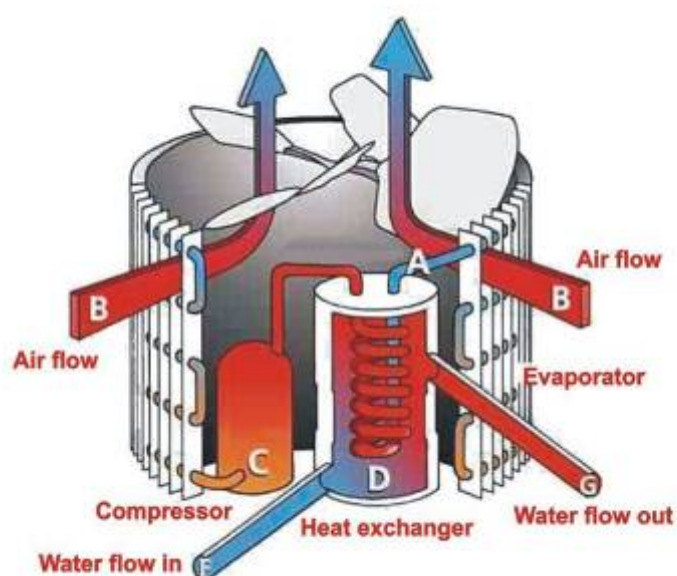


4. Principle

Swimming pool heat pump is a type of “transport device” that raises the temperature level of the heat that is freely available in the environment. It extracts stored solar heat from the environment —air and water— and transfers this, along with the operating energy, in the form of heat to the swimming pool system.

Heat cannot transfer from a cold body to a warm body on its own. Rather, it flows from a body with a higher temperature to a body with a low temperature (Second law of thermodynamics). For this reason, the heat pump must raise the temperature of the thermal energy extracted from the environment using high-grade energy to a level suitable for heating.

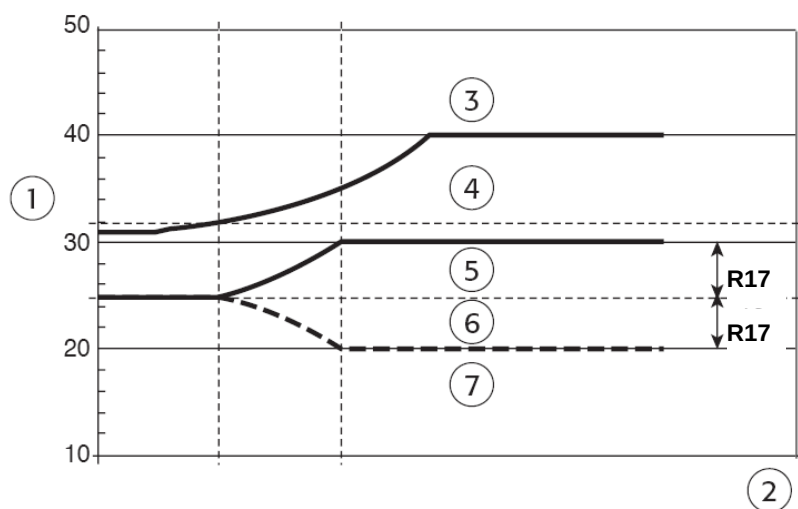
Heat pump work like a refrigerator- in other words, the same technology is applied but the function is reversed. It extracts heat from a cold environment which can then be used for heating.



5. Features

- ◆ Long work life
We use pure titanium coil for heat exchanger, it can resist corrosion and rust.
- ◆ High efficiency
The patent double channel design of heat exchanger, make the water heat exchange with water sufficient, ensure the unit gain the high heat exchange efficiency.
- ◆ Environment friendly
We use R410a green refrigerant, never damage the ozone layer.

- ◆ Wireless control (optional)
PHNIX swimming pool, you can use Iphone or Ipad remote control the unit.
- ◆ Ambient temperature compensation (optional)
The set point of the swimming pool heat pump can adjust automatic depend on ambient temperature.



Key:

1. Temperature;
2. Time;
3. External temperature;
4. Compressor starts temperature;
5. Positive compensation;
6. Set point;
7. Negative compensation;

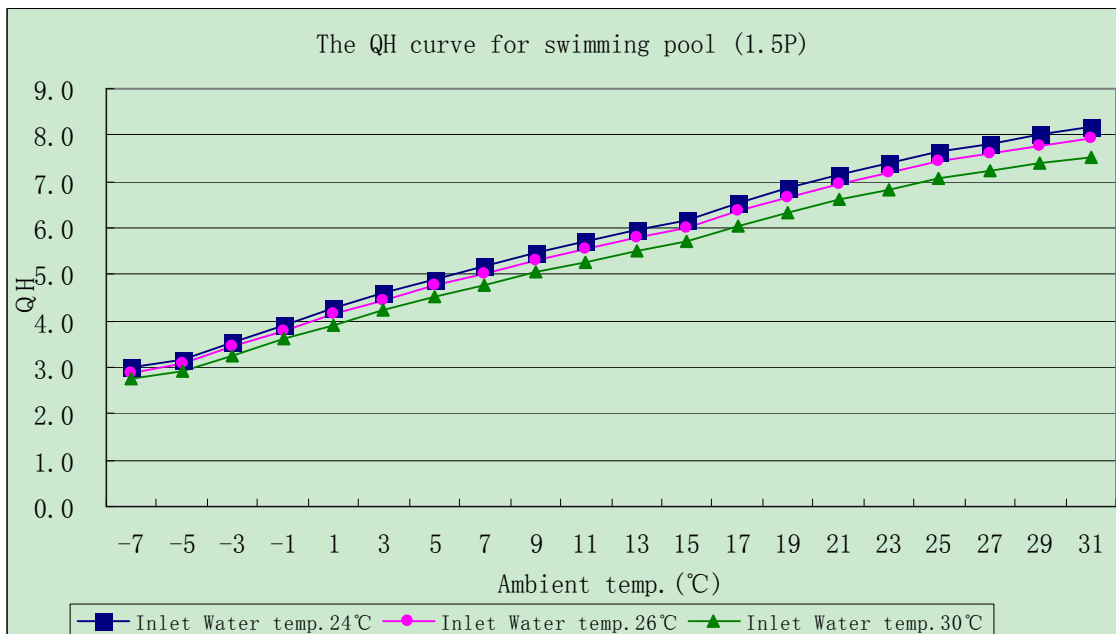
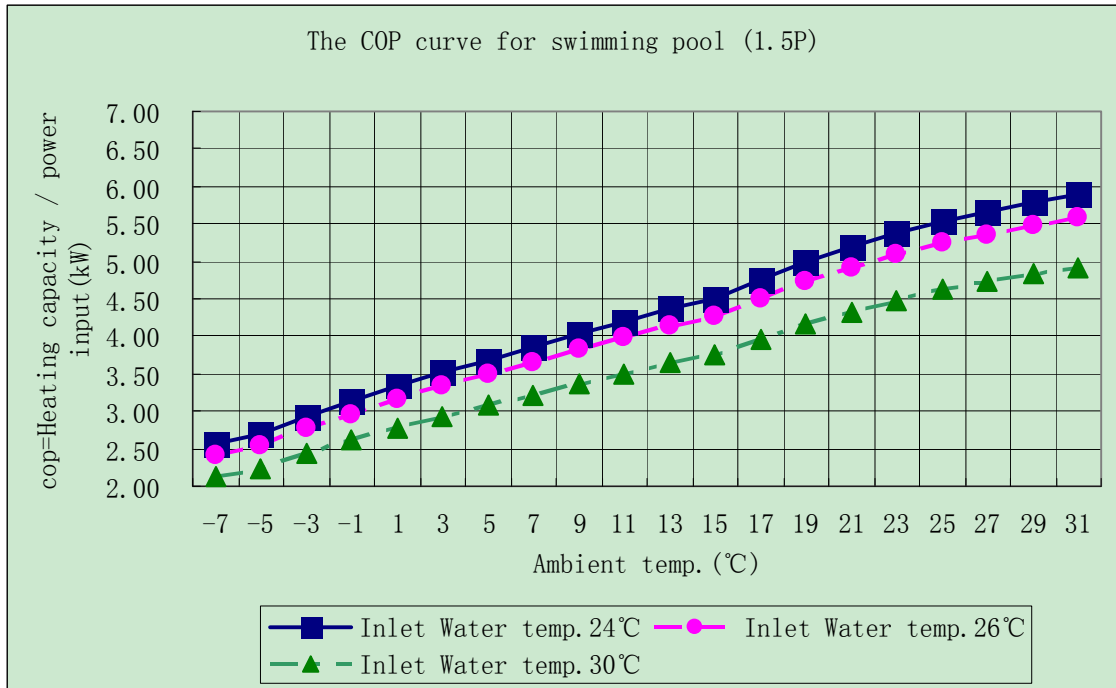
Section2 Performance and Specification

6. Specification

Model	PASRW	015	020	030	035	040
Heating capacity	kW	6.0	8.0	11.5	13.0	15.5
	BTU/h	20400	27200	39100	44200	52700
Heating power input	kW	1.41	1.87	2.48	2.88	3.48
COP	/	4.26	4.28	4.64	4.51	4.45
Heating current	A	6.4	8.5	11.3	13.1	15.8
Power supply		230V~/50hz	230V~/50hz	230V~/50hz	230V~/50hz	230V~/50hz
Compressor qty.		1	1	1	1	1
Compressor type	/	Rotary	Rotary	Rotary	Scroll	Scroll
Fan motor qty.	/	1	1	1	1	1
Fan motor power input	W	120	120	150	150	150
Fan rotation speed	RPM	850	850	850	850	850
Air flow direction		Horizontal	Horizontal	Horizontal	Horizontal	Horizontal
Noise	dB(A)	51	54	56	56	56
Water connection	mm	50	50	50	50	50
Water flow	m ³ /h	2.5	3.4	5	5.3	6.5
Water drop pressure	K Pa	10	10	12	12	12
Net dimentions(L/W/H)	mm	1025/455/660	1025/455/660	1140/470/875	1140/470/875	1140/470/875
Shipping dimension(L/W/H)	mm	1130/470/760	1130/470/760	1220/490/880	1220/490/880	1220/490/880
Net weight / shipping weight	kg	57/71	61/75	80/98	106/124	106/124
■ Value at +/- 5% under the following condition: Exterior temperature= 15°C(59°F) / HR=71% / Water inlet temperature = 26°C(78.8°F) / ΔT water 2°C(3.6°F).						
■Above information just for your reference. Please subject to nameplate on the unit.						

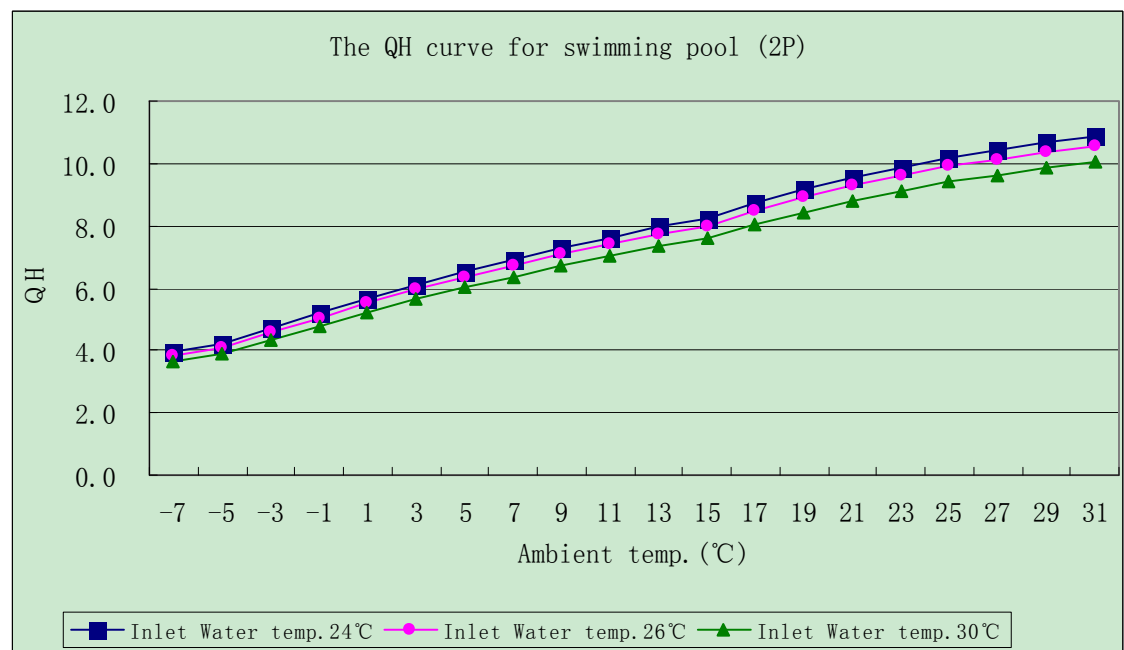
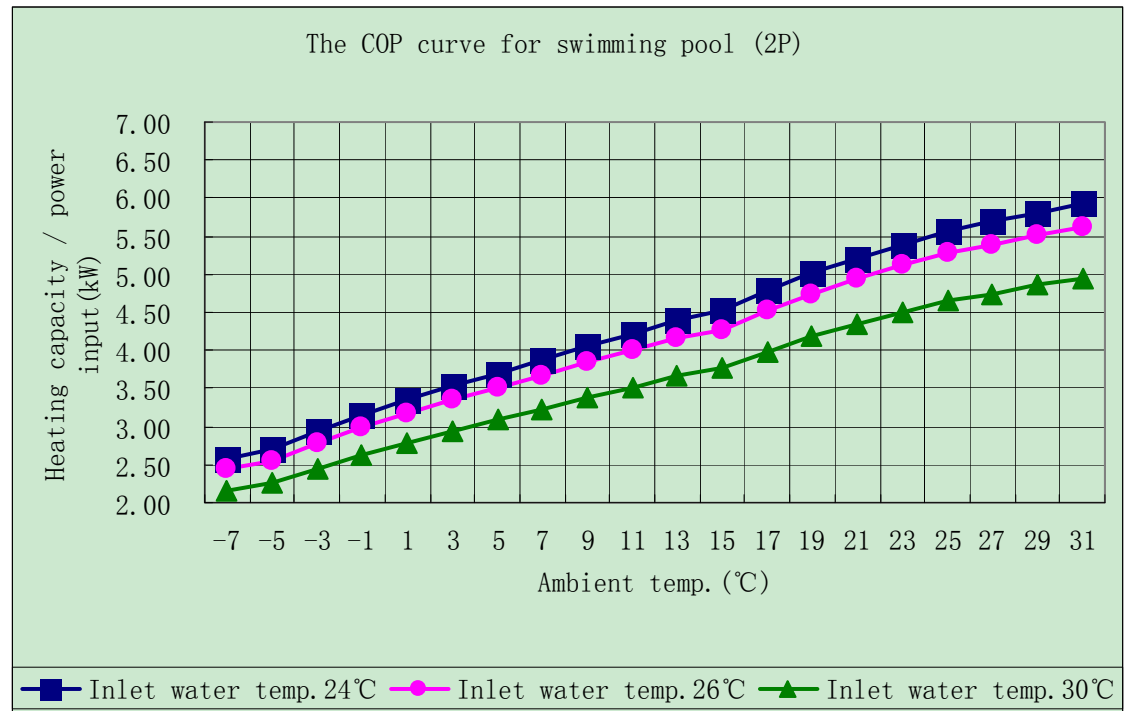
◆ Performance curve

performance curves PASRW015



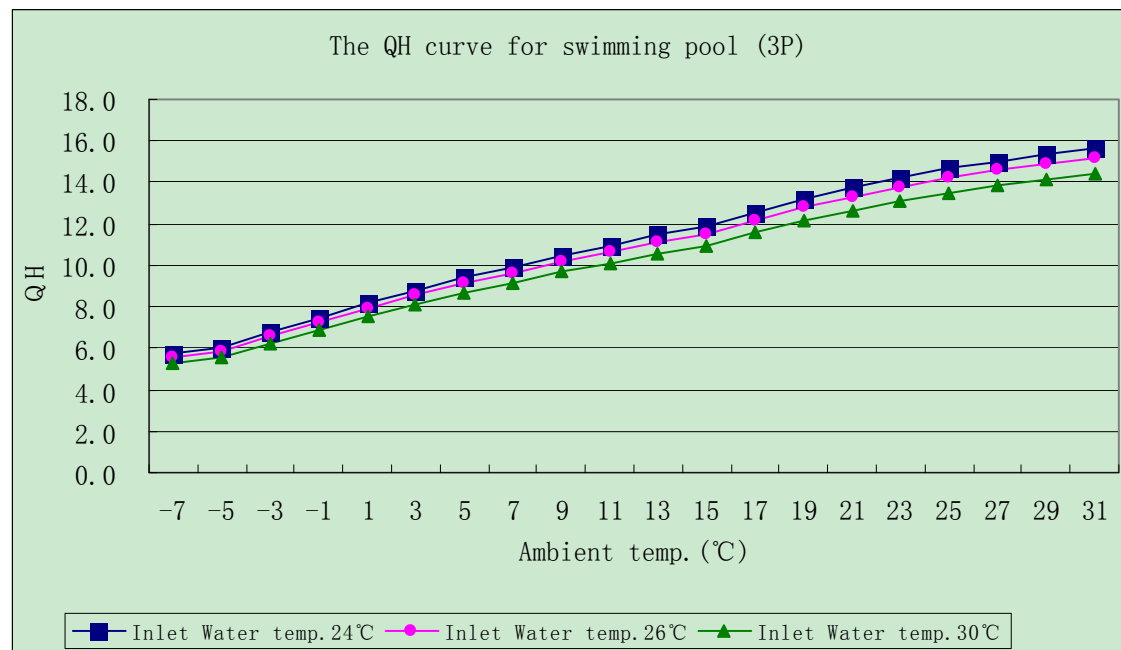
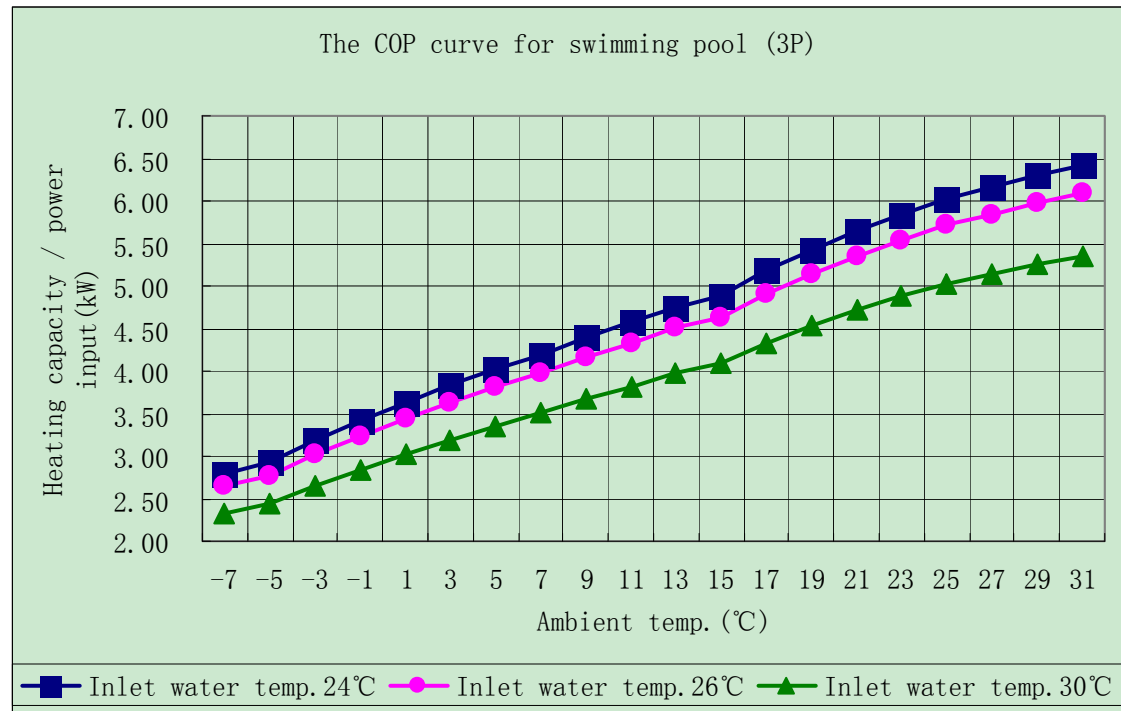
Swimming pool heat pump

Performance curves PASRW020



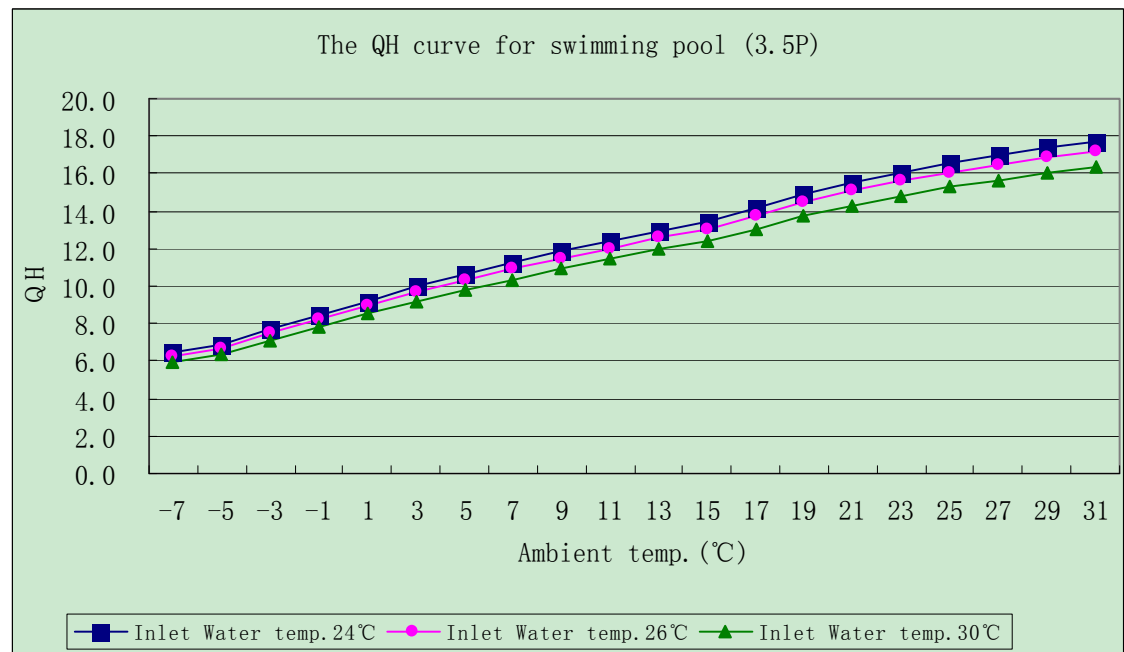
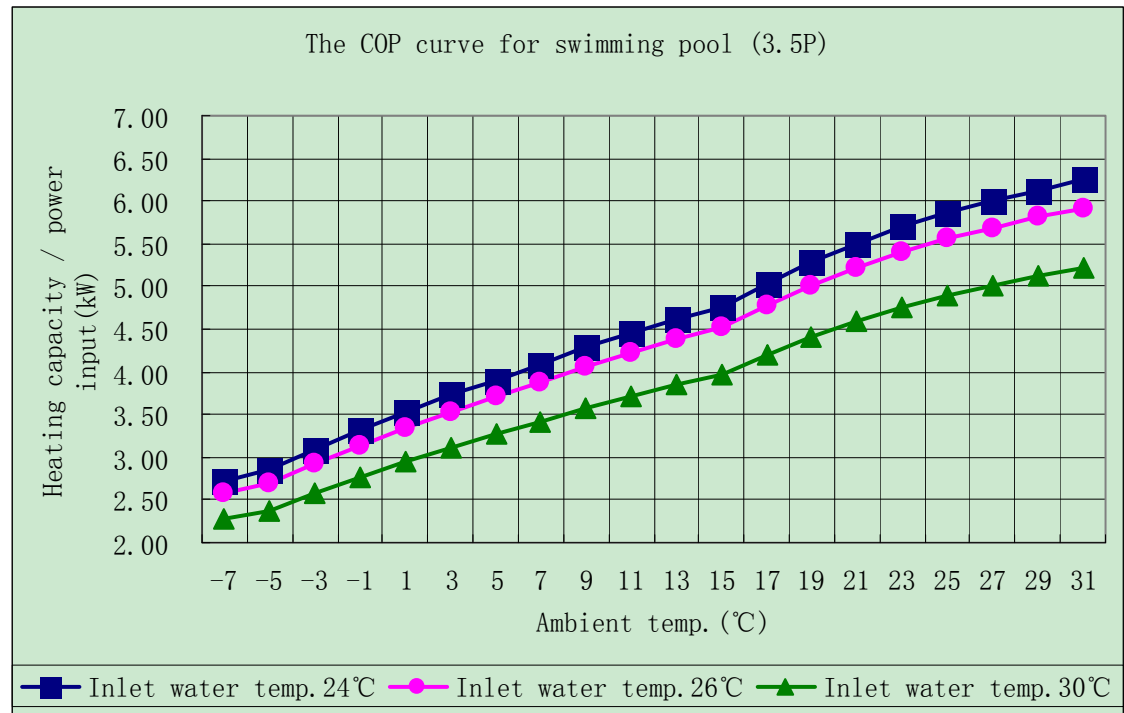
Swimming pool heat pump

Performance curves PASRW030



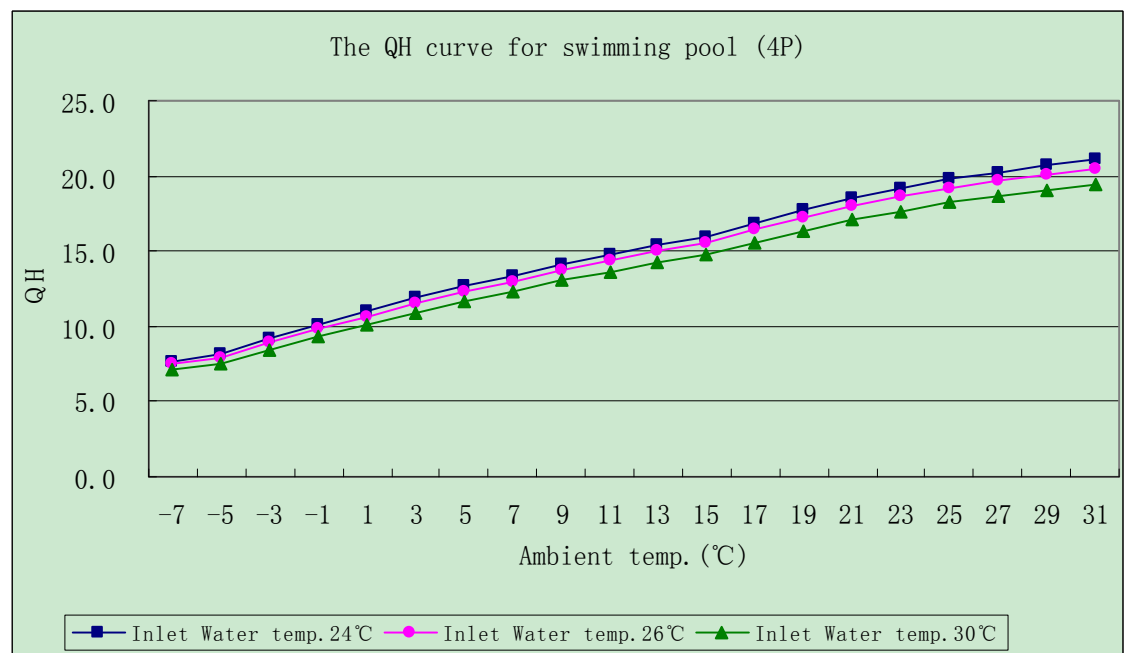
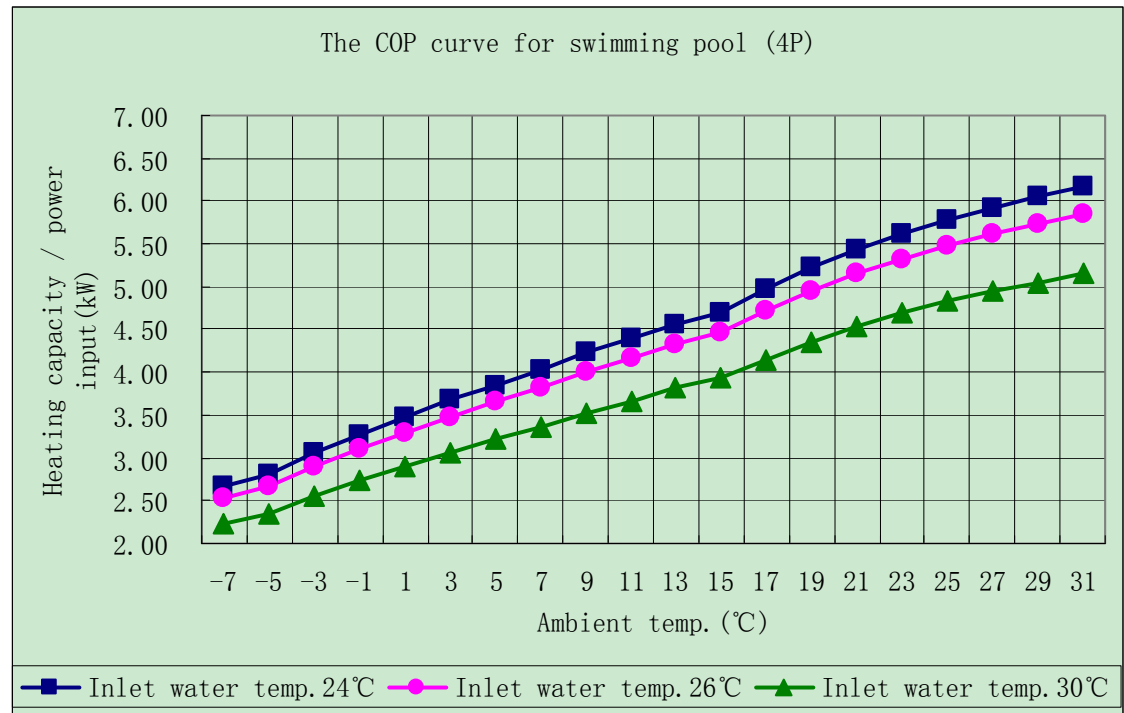
Swimming pool heat pump

Performance curves PASRW035



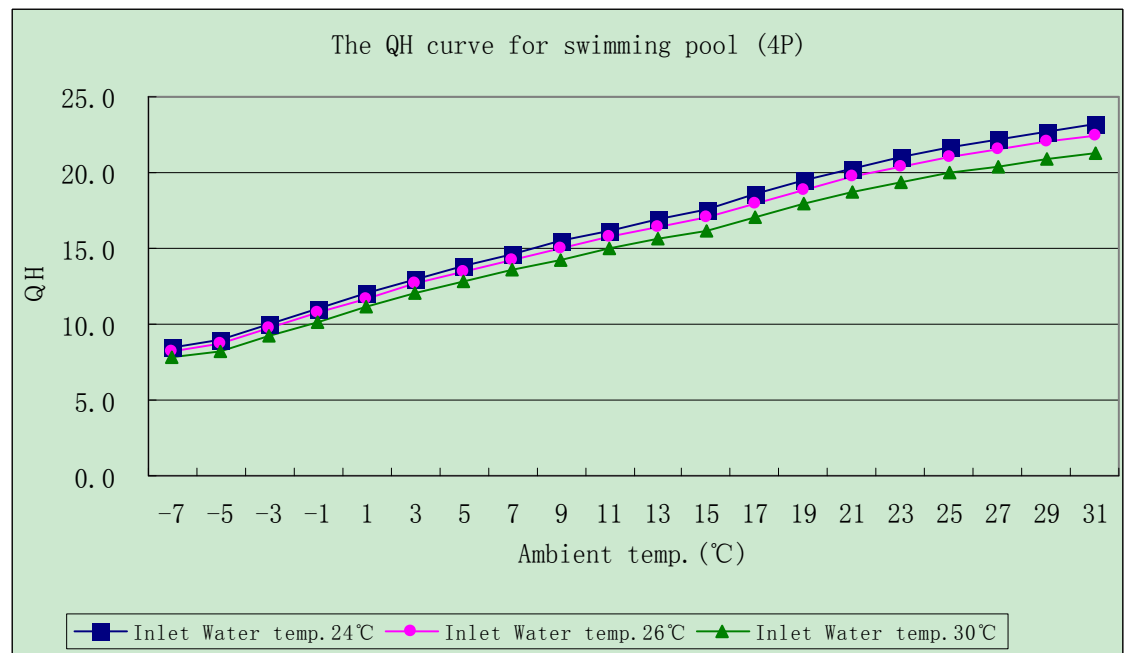
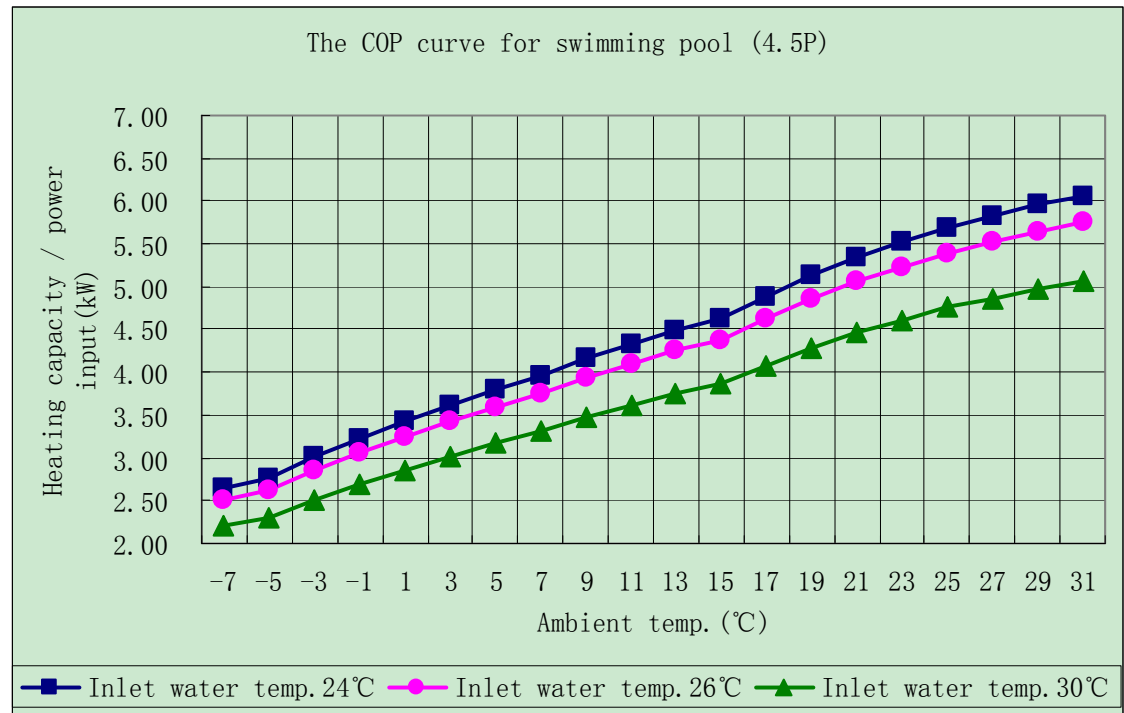
Swimming pool heat pump

Performance curves PASRW040



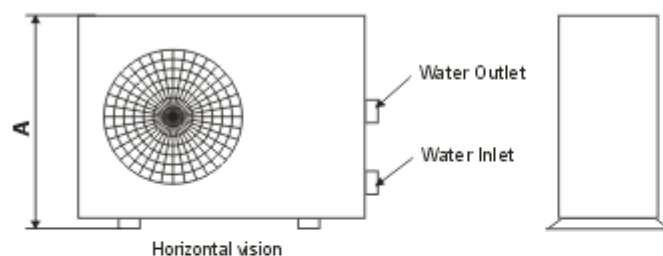
Swimming pool heat pump

Performance curves PASRW045



◆ Dimension

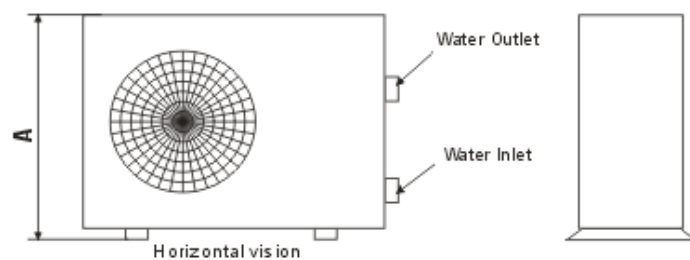
Model: PASRW015



unit: mm

TYPE SIZE \ PASRW	015
A	550
B	956
C	650
D	390
E	366

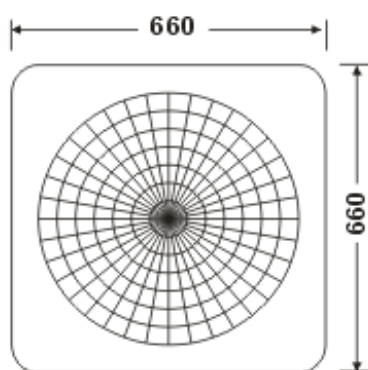
Model: PASRW020/030/040



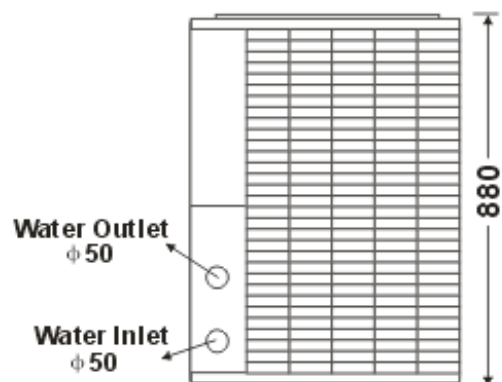
unit: mm

TYPE SIZE \ PASRW	020-030	040
A	650	850
B	1000	1120
C	700	760
D	420	470
E	400	440

Model: PASRW050S/060S



Vertical vision



Horizontal vision

7. Component instruction

- 1) Heat exchanger
 - a) Use own patent highly efficiency tube in shell heat exchanger.
 - b) We use Titanium pipe inside of heat exchange for resist corrosion.
 - c) Double coils inside for high efficiency.
 - d) Low water pressure drop.



Items	Specification	
	Titanium pipe	Outer shell
Resistance water pressure	4.5MPa	2.0MPa
Tightness of gas	3.2MPa	0.5MPa
Water pressure drop		
Water connection	/	PVC ϕ 50
Storage condition	-50~150°C	0~50°C
ROHS	Meet the requirements	

Section 3 Installation

8. Installation notice

To protect user him and others away from harms and dangers and correctly and safely use the hot water heat pumps, please carefully read the content here. PHNIX will not take any responsibility of personal dead, injury or machinery breakdown caused by any misuse and mistakes of non-compliance to regulation or signs in this manual.

MEANING OF WARNING AND CAUTION SIGN AND MARK

SIGN	MEANING
 WARNING	Misoperation (use) leads to personal dead or serious injury.
 CAUTION	Misoperation (use) leads to personal serious injury or machine breakdown.

SIGN	MEANING
	It means prohibition and the forbidden contents are shown by graphics or words around.
	It means compulsory execution and the forbidden contents are shown by graphics or words around.
	It means notice (including warning) and the detailed contents are shown by graphics and words around.

WARNING

SAFETY WARNING		The swimming pool heat pump unit must be earthed effectively.
		Ask your supplier for installation of the swimming pool heat pump units. Incomplete installation performed by yourself may result in water leakage, electric shock, or fire.
		The unit must be securely fixed, or noise and shaking will be resulted.
SAFETY NOTICE		The unit can not be installed at the place where combustible gas is easy to leak. Once the combustible gas leaks, the place around unit is probably fired.

Swimming pool heat pump

		Confirm whether the unit is installed with leakage protective switch. If not, it can cause personal electric shock or fire.
		The appliance shall be installed in accordance with national wiring regulations.
USAGE WARNING		Do not insert fingers, rods or other objects into the air outlet. When the fan is rotating at high speed, it will cause injury.
		In order to avoid electric shock, fire or injury, if any abnormality is detected, such as smell of fire, turn off the power supply and call your supplier for instructions.
		Never touch the internal parts of the controller.
MOVE AND MENDIGN WARNING		When hot water heat pump is moved and re-installed, clients have to let distributors or professional technicians to do it. If the unit is not well installed, electric shock, fire, injure and water leakage would probably occur.
		Absolutely no mending by user himself, or this will probably lead to electric shock and fire.
		Ask your supplier for the repair and maintenance. Incomplete repair and maintenance may result in water leakage, electric shock or fire.
USAGE NOTICE ITEMS		Before cleaning, be sure to stop the operation and turn the breaker off or pull out the power cord. Otherwise, an electric shock and injury may be caused.
		Never replace a fuse with that of wrong rated current or other wires when a fuse blows out. Use of wrong wire or copper wire may cause the unit to break down or a fire.
		Never use a flammable spray such as hair spray, lacquer paint near the unit. It may cause a fire.
		Do not dispose this product as unsorted municipal waste. Collection of such waste separately for special treatment is necessary.
		Do not use the swimming pool heat pump for other purposes.
		In order to avoid injury, do not remove the fan guard on the unit.
		Do not operate swimming pool heat pump with a wet hand. An electric shock may be caused.

9. Deposition and transportation

As a rule, the unit is to be stored and/or transported in its shipping container in upright position and without water charge. For a transport over short distance, and provided due care is exercised, an inclination angle of up to 30 degree is permitted. Both during transport and storage, ambient temperatures of **- 20 to +70°C** are permissible.

a) Transport using a forklift

When transported by a forklift, the unit must remain mounted on the pallet. To prevent any damage, the unit must be placed on a level surface!

b) Manual transport

For the manual transport, the wooden pallet can be used for bottom part. Using ropes or carrying straps, a second or third handling configuration is possible. With this type of handling, care must be taken that the max. Permissible inclination angle of 60 degree is not exceeded.

10. Inspecting and Handling the Unit

After delivery, the package should be checked and any damage should be reported immediately to the carrier claims agent.

When handling the unit, take into account the following:

- ◆  Fragile, handle the unit with care.
- ◆  Keep the unit upright in order to avoid compressor damage.
- ◆  Keep the unit dry
- ◆  Units can be stocked up, it can be 2/3/4 layers according to unit model
- ◆ Move this unit with original package.
- ◆ When lifting the unit, always use protectors to prevent belt damage and pay attention to the balance of the unit's gravity.

11. Unpacking

When unpacking the unit, take into account the following:

- Check this unit appearance
- Check this unit screw fastening
- Check this unit bar code, nameplate and others label
- Check the unit accessories

Accessories

Check whether the following assemblies are complete.

Name	Quantity	Note	Remarks
Manual	1	—	/
Damping rubber	4	—	/
Pipe connection	2	—	/
Wire controller	1	—	LED / LCD

12. Installation information

- Enough space for installation and maintenance shall be preserved.
- The air outlet should be free from obstacles and strong wind.
- The bearing surface should be flat, able to bear weight of the unit and suitable for installing the unit horizontally without increasing noise or vibration.
- No flammable gas is leaked nearby.
- It is convenient for piping and wiring.

1) Installation position

Installing the equipment in any of the following places may lead to malfunction of the equipment (if it is inevitable to consult the supplier):

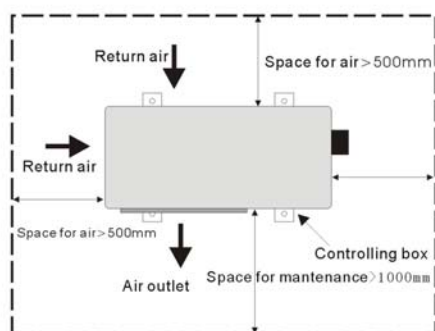
- ◆ The site contains mineral oils such as cutting lubricant.
- ◆ Hot spring area where corrosive gases exist, e.g., sulfide gas.
- ◆ Factories where the power voltage fluctuates seriously.
- ◆ Place where strong electromagnetic waves exist.
- ◆ Place where flammable gases or materials exist.
- ◆ Place where acid or alkali gases evaporate.
- ◆ Other special environments.

2) Precautions before installation

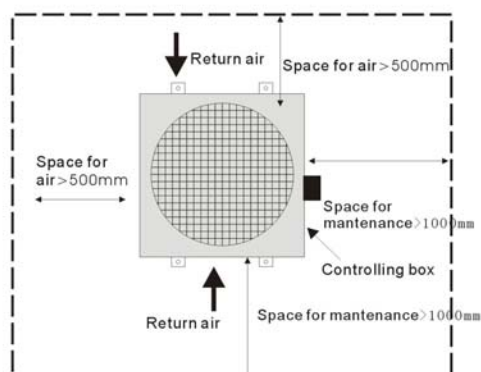
- ◆ Decide the correct way of conveying the equipment.
- ◆ Try to transport this equipment with the original package.
- ◆ If the unit has to be installed on a metal part of the building, electric insulation must be installed, and the installation must meet the relevant technical standards for electric devices.

3) Installation space

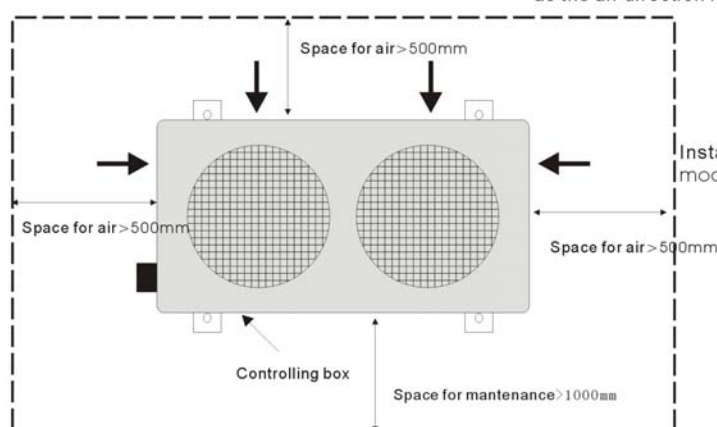
- ◆ Before installing the unit, reserve the space of maintenance.
- ◆ The installation space should be convenient to install.
- ◆ The installation space should with good ventilation.
- ◆ The installation space should not to against air flow directly.
- ◆ There must be drainage channel around the unit for condensing water.
- ◆ The installation location should be better near power supply.
- ◆ Try to close the swimming pool for the convenience of piping.
- ◆ Parallel connection must be carried out when install several units together.



Installation space for side discharge units:
Model: PASRW010/020/030/040/050S/060S



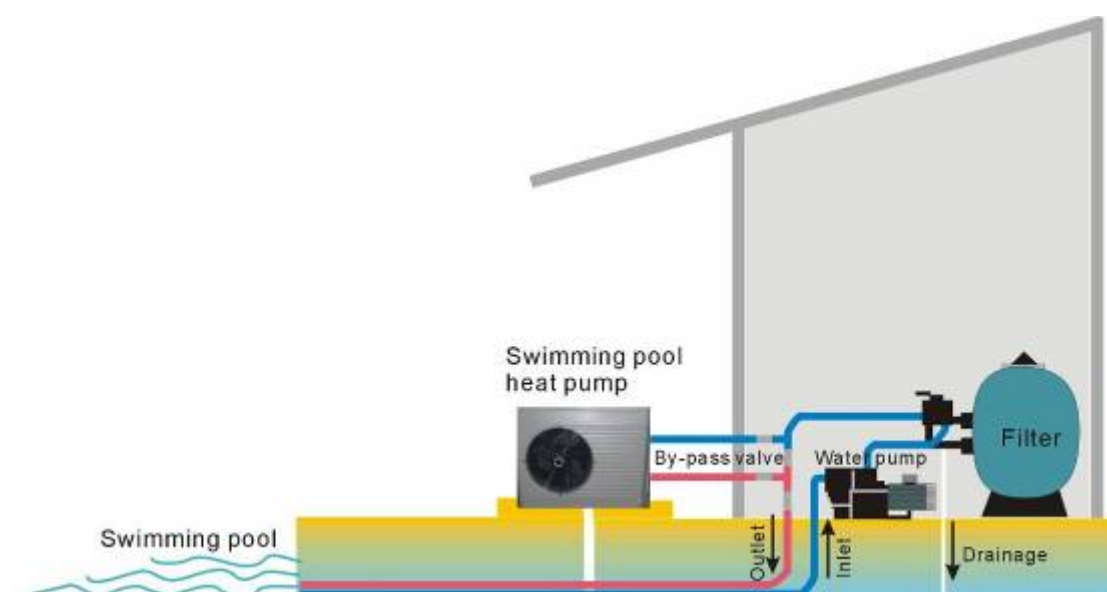
Installation space for top discharge unit:
Model: PASRW060_V/060S_V/080S_V
there must be no obstacle on the top of these units
as the air direction is vertical



Installation space for side discharge units:
model : PASRW100S_U/130S_U

13. Hydraulic system

◆ Installation sketch map



◆ Hydraulic system

The connection of water system is simple; we just need the PVC pipes, glue for PVC, and saw to finish the connection. The factory provides the standard PVC connectors for every unit.

Please connect the water pipe according to instructions below:

Recognize the specification, length and quantity of the PVC pipes; prepare tools;

Cut the PVC pipes with saw; make a smooth round end; make sure that the pipe be cut vertically;

See below picture:



The end of the PVC pipes must be thoroughly cleaned by dry cloth; the oil or smear at the PVC end must be cleared by dry cloth with acetone, if there is any;

Make a try to insert one pipe into another one and check the depth and fitness; Mark at the pipe and then brush the PVC glue at the surface of the small pipe and the interior of the wide pipe quickly;

Insert one pipe into the big one immediately to the marked place; keep it for 30 seconds (when the diameter of the pipe is less than 63 mm), adjust it accordingly to a good appearance;

Clean the glue on the PVC when it is finished.

Attention:

Water can not be conducted to PVC pipe at least 12 hours after the pipe is stuck.

Please connect the pump, filter, heat pump and other accessories with removable coupling for convenience of maintenance and change.

14. Electrical Wiring

- Before wiring, confirm the rated voltage of the unit as shown on its nameplate, then carry out the wiring closely following the wiring diagram;
- It is not allowed to change the wire connection of the unit by user; wire connection must be made by qualified electrician; make sure the earthing is well, any change on the way of earthing is not allowed.
- Each wiring connection must be done in accordance with system diagram. Wrong wiring may cause unit to misoperate or become damaged.
- Do not allow wiring to touch the refrigerant tubing, compressor, or any moving parts of the fan. Unauthorized changes in the internal wiring can be very dangerous. The manufacturers will accept no responsibility for any damage or misoperation that occurs as a result of such unauthorized changes.
- Regulations on wire diameters differ from locality. For filed wiring rules, please refer to your local electrical codes before beginning. You must insure that installation complies with all relevant rules and regulation.
- Please make the electrical connection according to the instructions below;



A) take out the screw on the protection cover and then take it off;



B) Then you can see the power port



C) Connect power wiring to the terminal according to the marks of L/N/⊕ (single) or R/S/T/N/⊕ (three phase);



D) The wire controller has been fixed on the unit body.

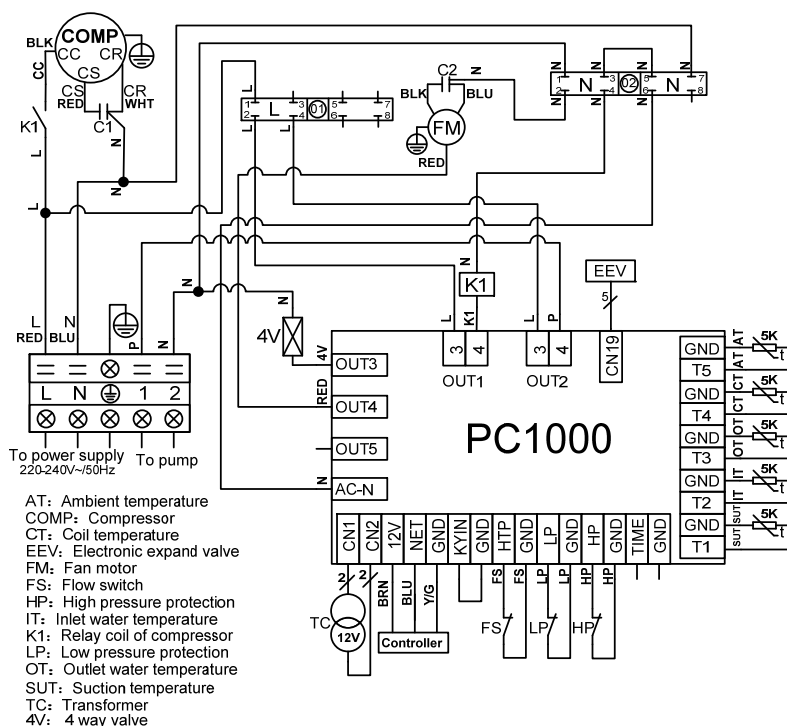
Notes: Please connect wiring to power switch at the final step to protect personal safety.

Swimming pool heat pump

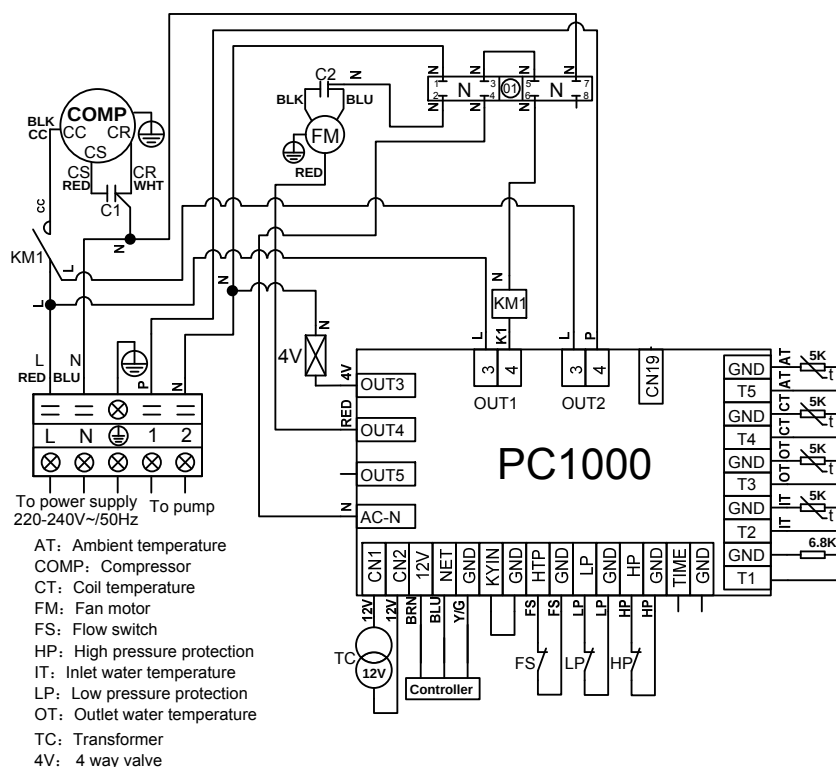
● Electric wiring diagrams

PC1001

Model: PASRW020-P

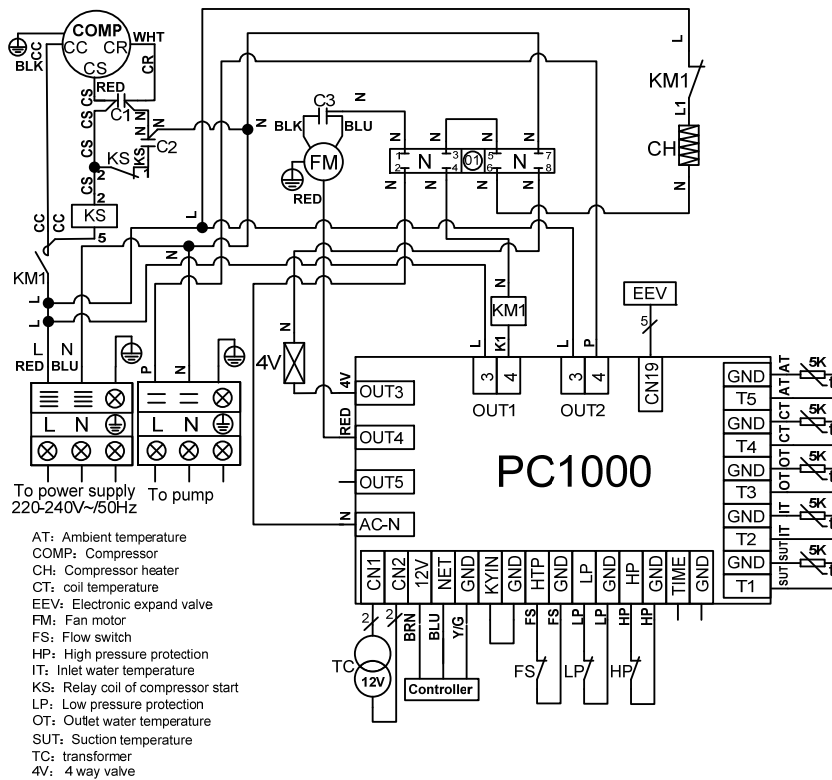


Model: PASRW030-P

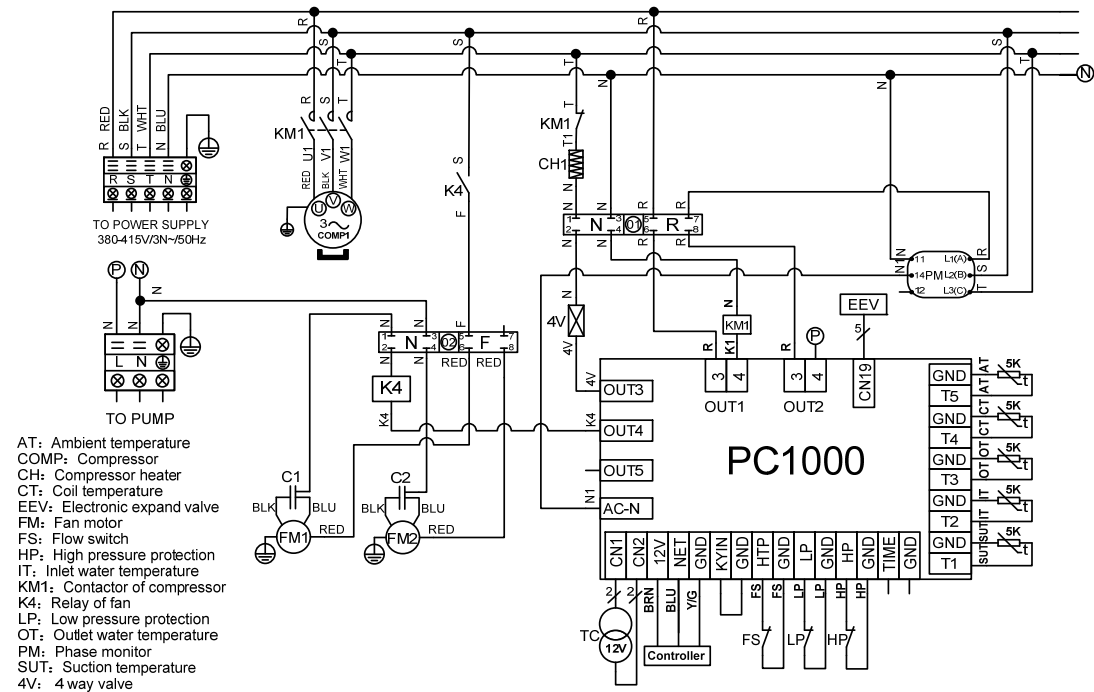


Swimming pool heat pump

Model: PASRW040-P



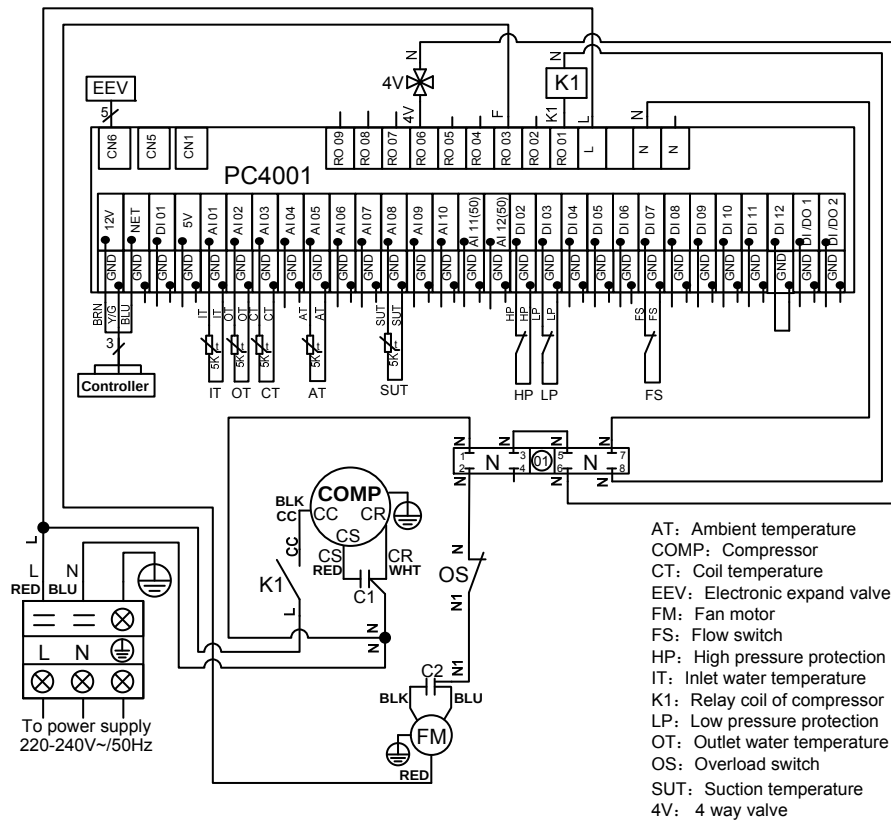
Model: PASRW050~060S (-U)-P



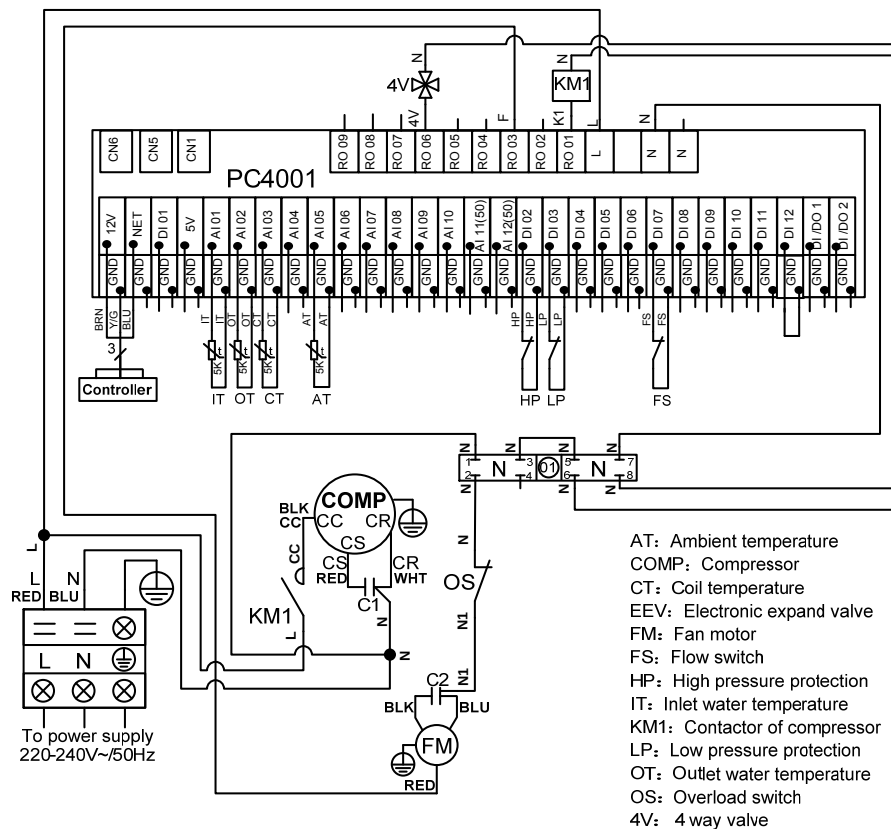
Swimming pool heat pump

PC4001

Model: PASRW020-P

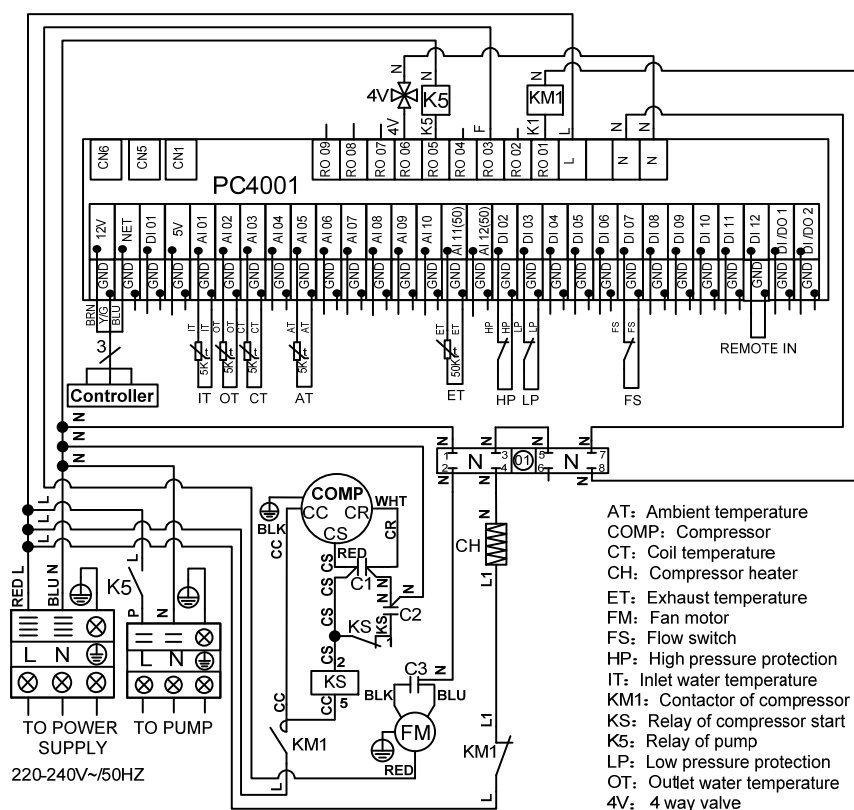


Model: PASRW030-P

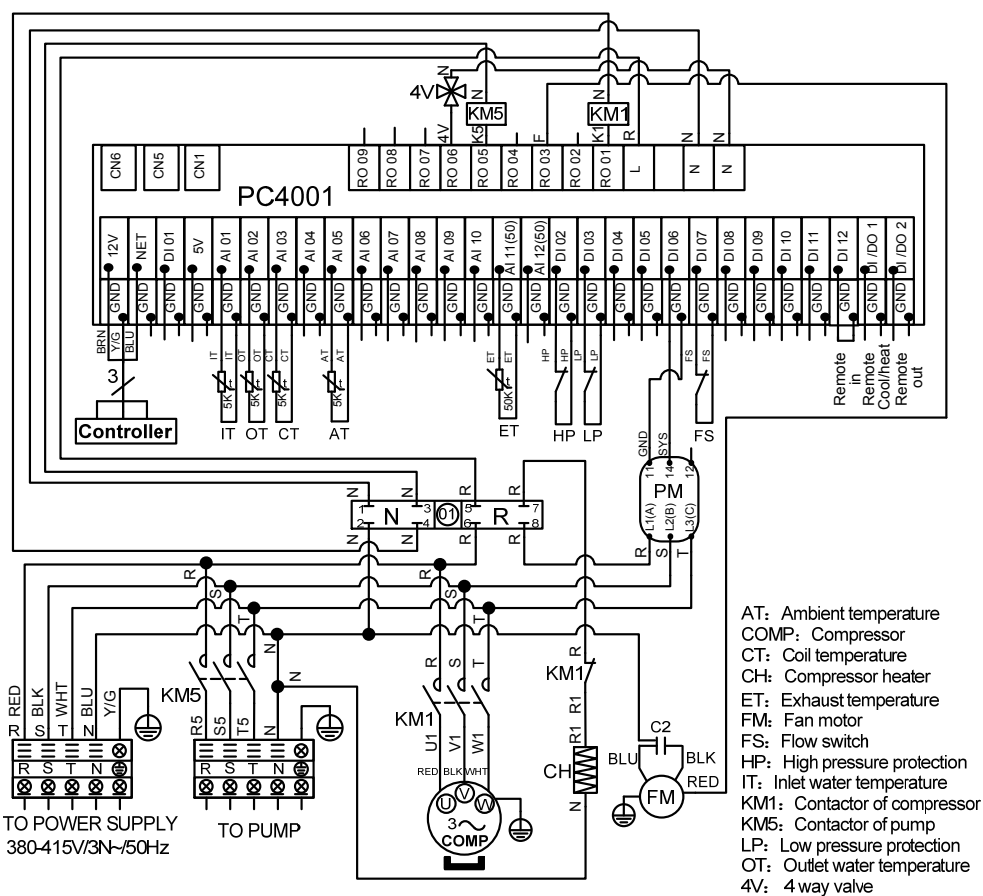


Swimming pool heat pump

Model: PASRW040 (-U)-P



Model: PASRW050~060S (-U)-P



15. Trial Running

When the unit runs for the first time, please follow steps below

- 1) Start the water pump to start the water circulation; check the water flow volume; make sure that the real water volume is not lower than the rated water flow volume that on the name plate by 60%;
- 2) Electrify the unit, and when the unit get electric power, press key to start the unit;
- 3) The fan, 4-way valve and compressor will start in order; examine the normal air temperature and the outlet air temperature from the units 40 seconds latter; and the latest one will be higher than the first one by 5°C-12°C;
- 4) Check the inlet water temperature and the outlet water temperature; it can be showed at the wire controller; the outlet temperature will be higher than the inlet water temperature by 1°C-2°C;
- 5) When the unit runs for 1 hour, we can feel that the water temp. in the pool goes higher; If the ambient temperature is higher than 30°C, the fan will run intermittently; This is a protection on the unit system;
- 6) When the water pump lose power and stop running, the unit will stop automatically because there are flow switch inside the unit; and the unit will start automatically when the water pump get power and start to run next time;
- 7) Please shut off the power supply for the unit and pump and, drain the water out from the unit if the unit will not be used for a long time.

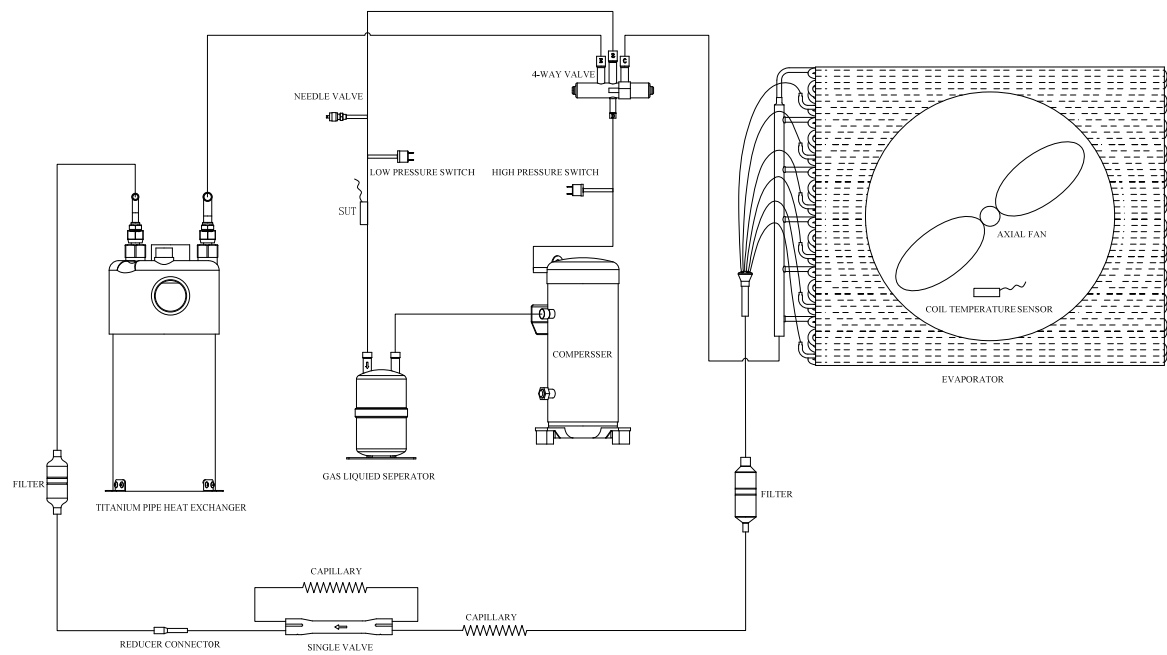
16. Thermal insulation for swimming pool



Attention: When ambient temperature is too low, the heat loss rate for swimming pool will be speed up. So you can put swimming pool cover on it to prevent heat loss. (As in the picture above)

Swimming pool heat pump

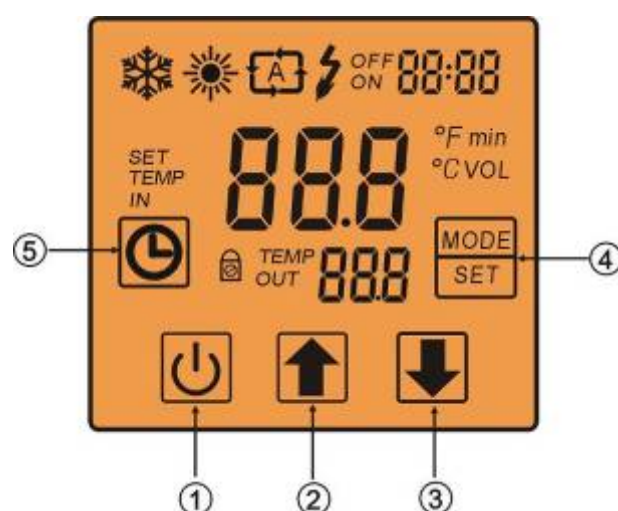
- System diagrams



Section 4 PC1001 Controller

17. Controller Operation

◆ Control panel explanation



① On/Off key

Press this key to switch on or off; (hold on 0.5 s)

② Up key

- a) Select higher item inside programming environment;
- b) Increase parameter value

③ Down key

- a) Select lower item inside programming environment;
- b) Decrease parameter value;

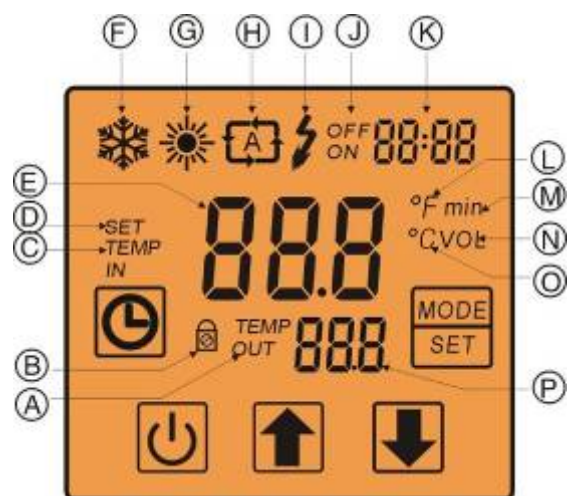
④ Mode key

- a) Select mode and set parameter.
- b) Release more factory parameters; (press for 10 s)

⑤ Clock key

Set date and timer

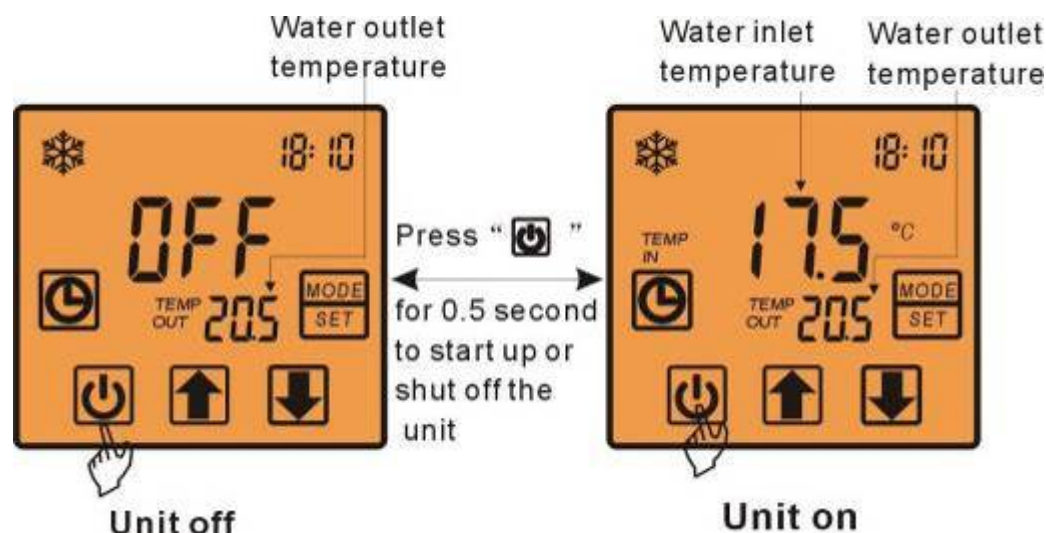
◆ Display Explanation



Symbol	Explanation	Symbol	Explanation
A	Water outlet	I	Electrical heater(reserved)
B	Lock	J	Timer on/ off
C	Water inlet	K	Clock
D	Temperature setting	L	Fahrenheit
E	Water inlet temperature	M	Minute
F	Cooling	N	Flow
G	Heating	O	Centigrade
H	Automatic	P	Outlet water temperature

◆ Operation Instruction

a) On-off operation



When the unit is on, press "on/off" key for 0.5 seconds to start it up;

When the unit is off, press "on/off" key for 0.5 seconds to shut it down.

b) Mode selection

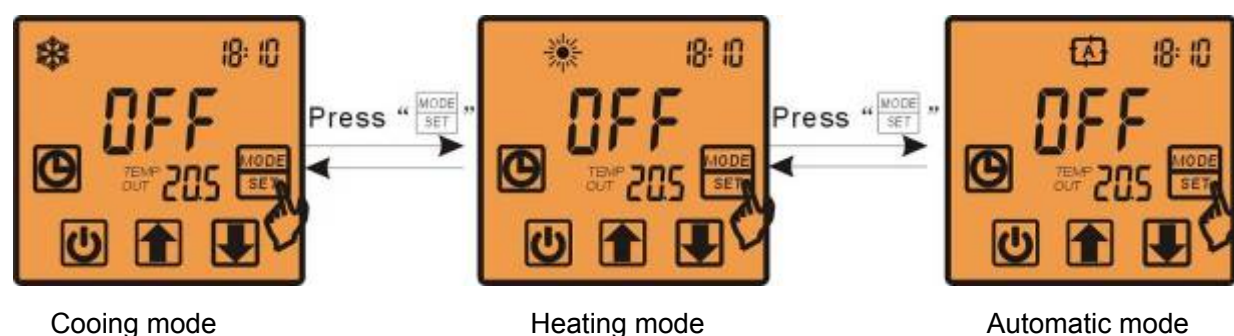
The unit has three operation modes:

Cooling mode, Heating mode, Auto heat pump mode.

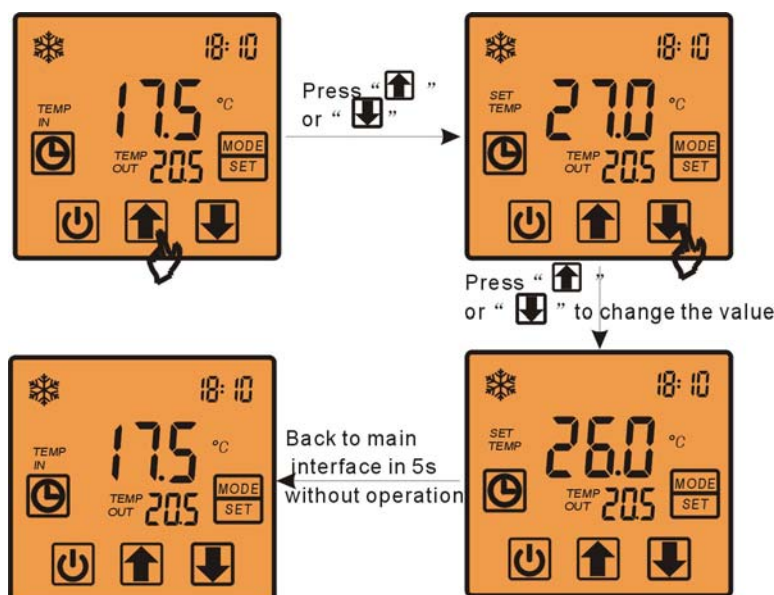
When parameter "H01=0", there is only "cooling mode" to choose;

When parameter "H01=1", Press "mode" can select modes of heating, cooling, automatic;

When parameter "H01=2", there is only "heating mode" to choose.



c) Target temperature Setting



- Attention:**
- 1) Press "mode" to save parameter setting and back to main interface.
 - 2) Press "on/off" can not save parameter setting but back to main interface
 - 3) If there is no operation for 5 seconds, the system will remember parameters setting and back to the main interface.

d) Parameter Setting

The parameters are divided into 2 different types, according to their level of access by the user (password) and their function.

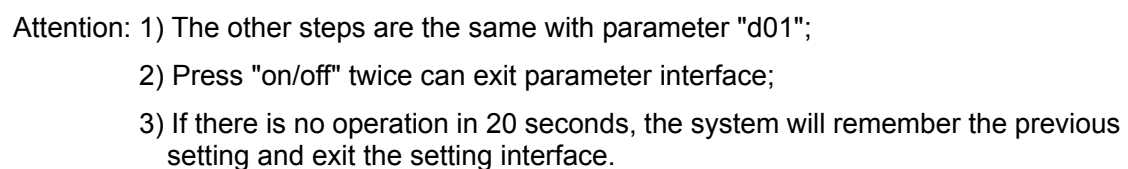
Factory parameters

Accessible with the 66 "factory" password, allow the configuration of all the unit parameters.

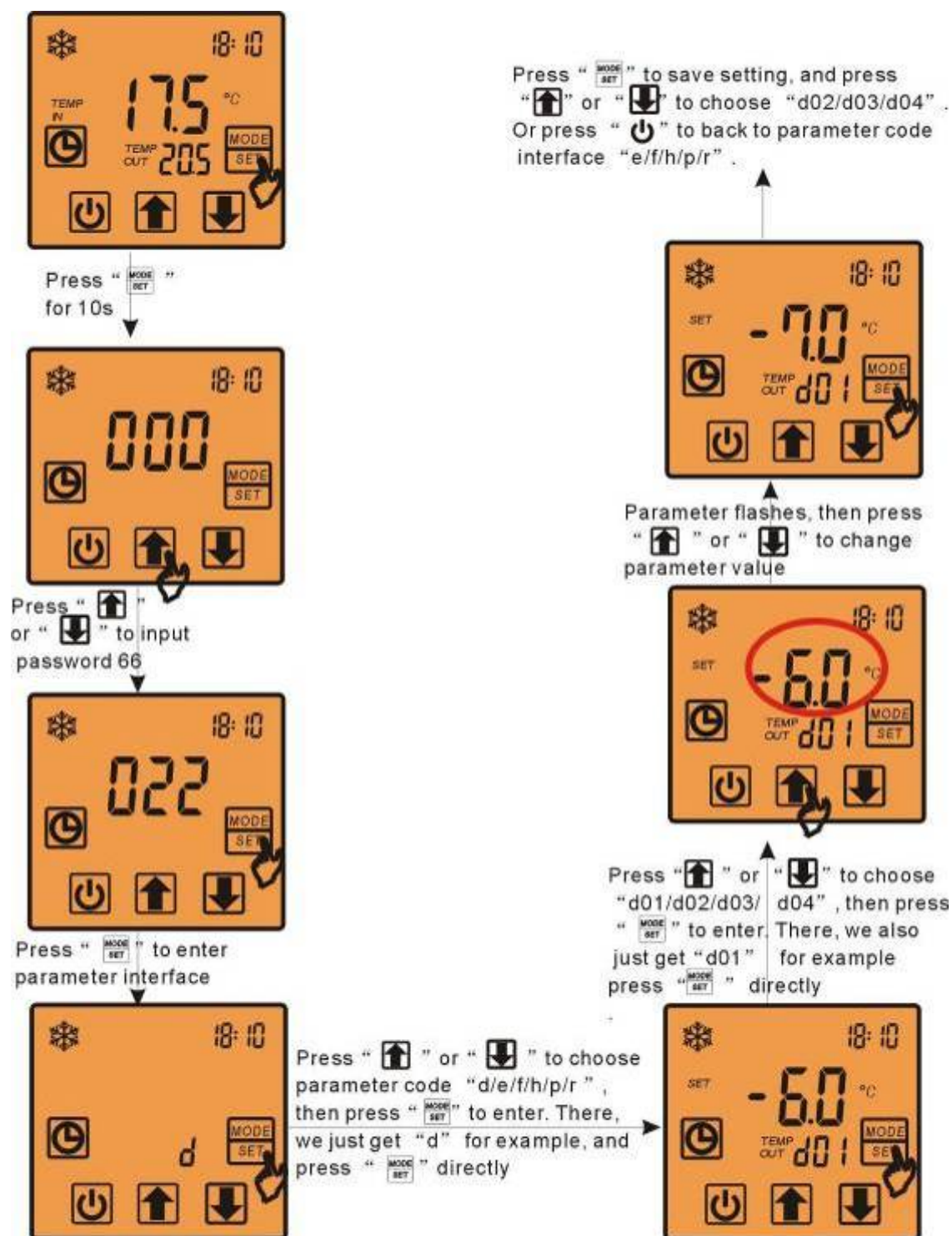
User parameters

Accessible with the 22 "User" passwords, allow the configuration of the parameters that typically can be set by the user.

level	Level name	password
U	User	22
F	Factory	66



User Parameter Setting

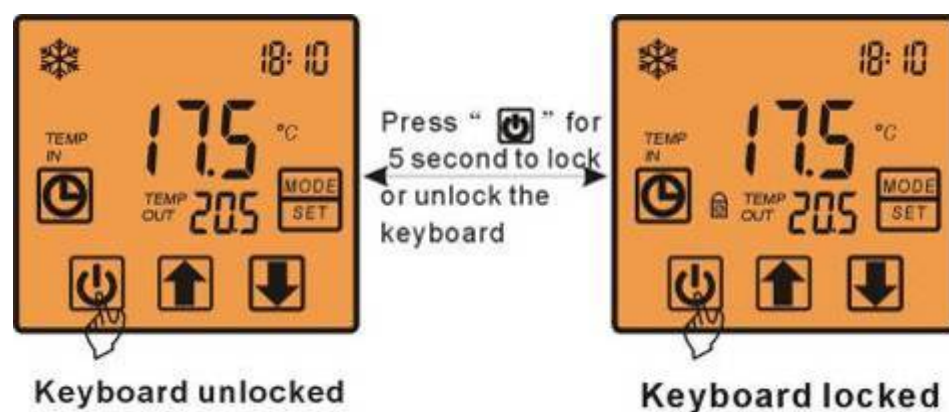


Attention: 1) The other steps are the same with parameter "d01";

2) Press "on/off" twice can exit parameter interface;

3) If there is no operation in 20 seconds, the system will remember the Previous setting and exit the setting interface.

e) Keyboard lock and unlock

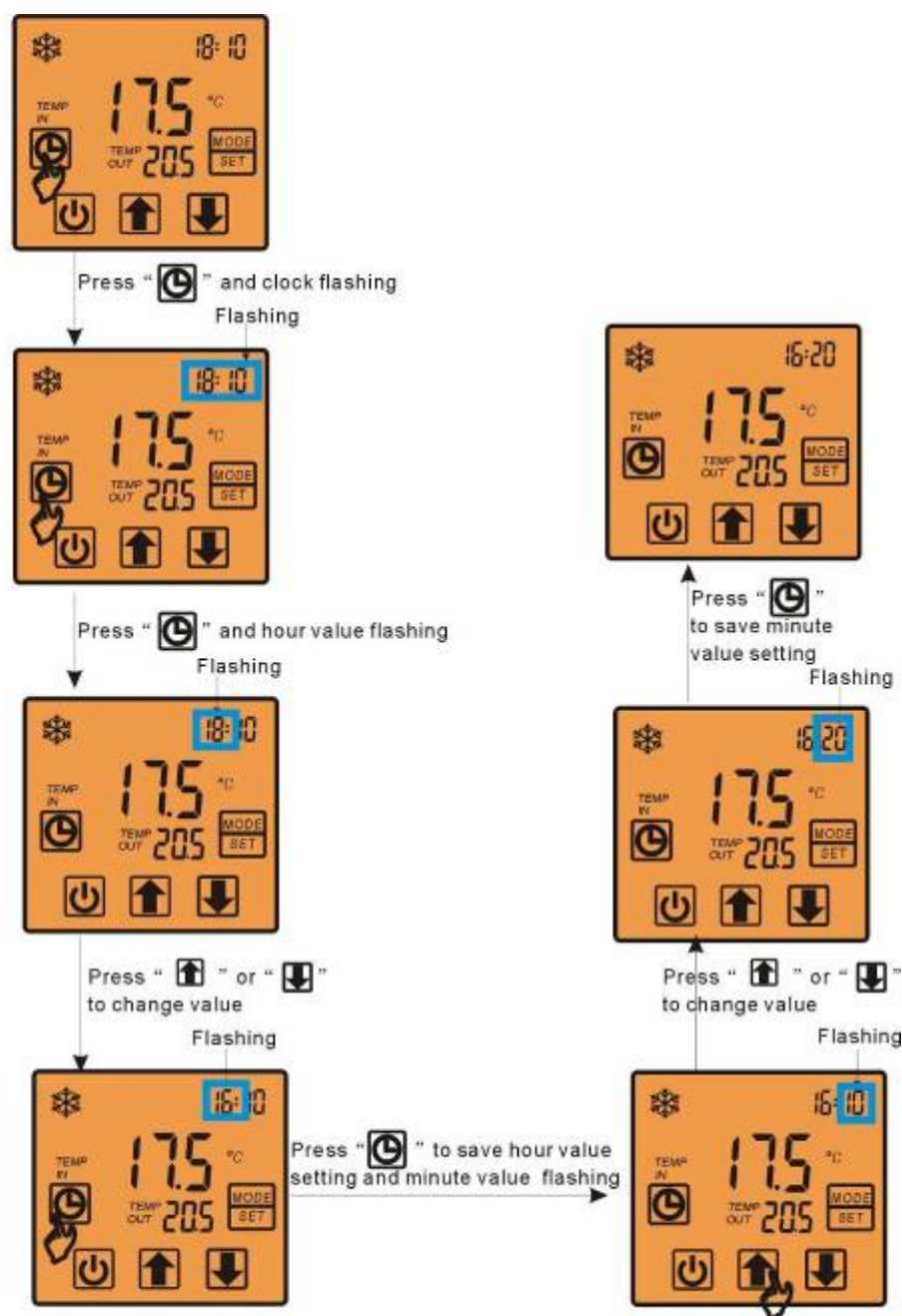


In on/off interface, press “on/off” for 5 seconds, the keyboard will be locked.

When the keyboard is locked, press “lock” for 5 seconds, the keyboard will be unlocked.

Attention: When the unit is in alarming state, the keyboard lock can be removed automatically.

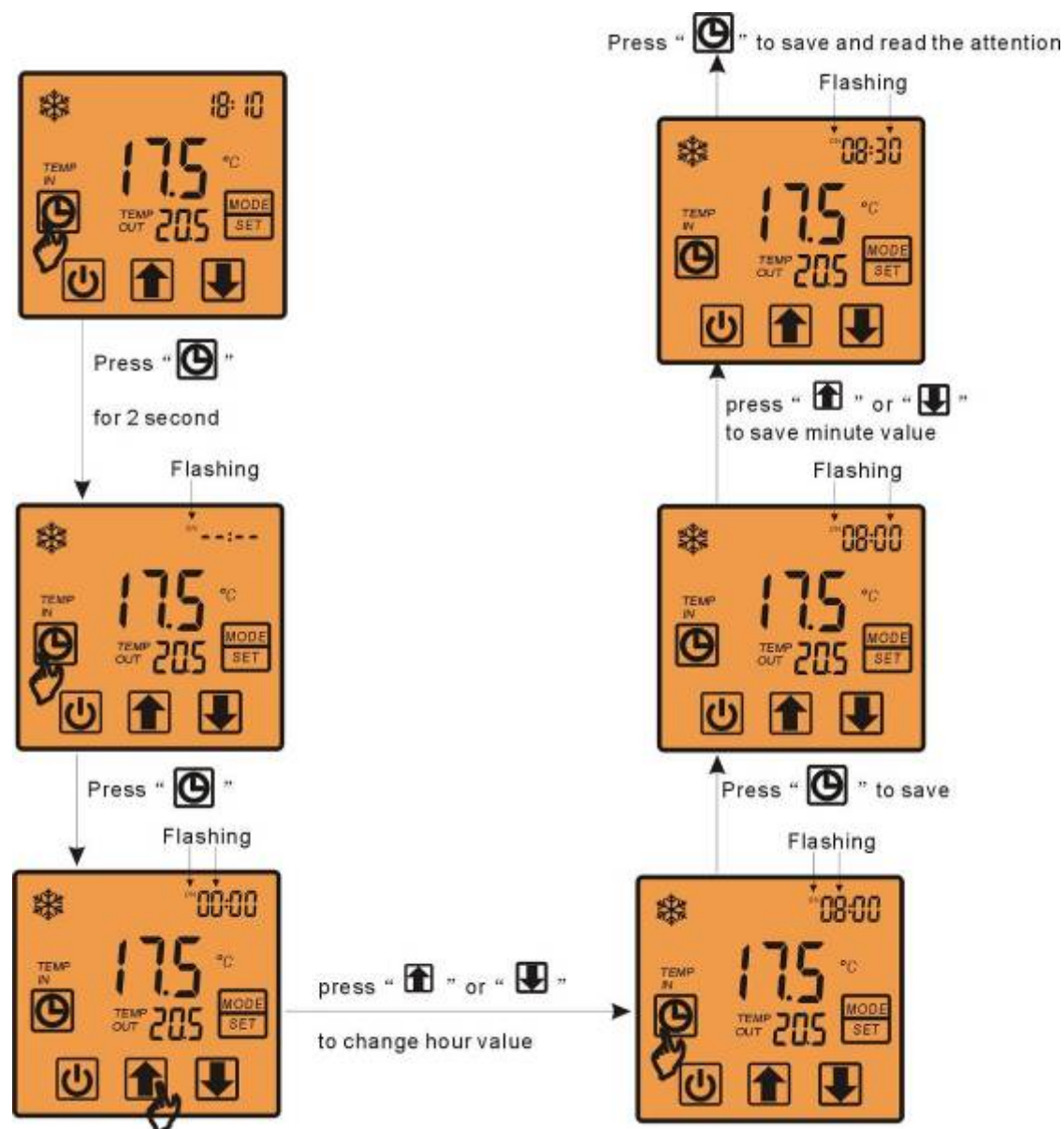
f) Clock setting



- Attention:
- 1) Press "clock" can save clock setting parameter and back to main Interface.
 - 2) Press " on/off" can not save setting parameter and back to main interface.
 - 3) If there is no operation for 5s, system will remember parameter setting and back to the main interface.

g) Timer on and off setting

Timer on setting



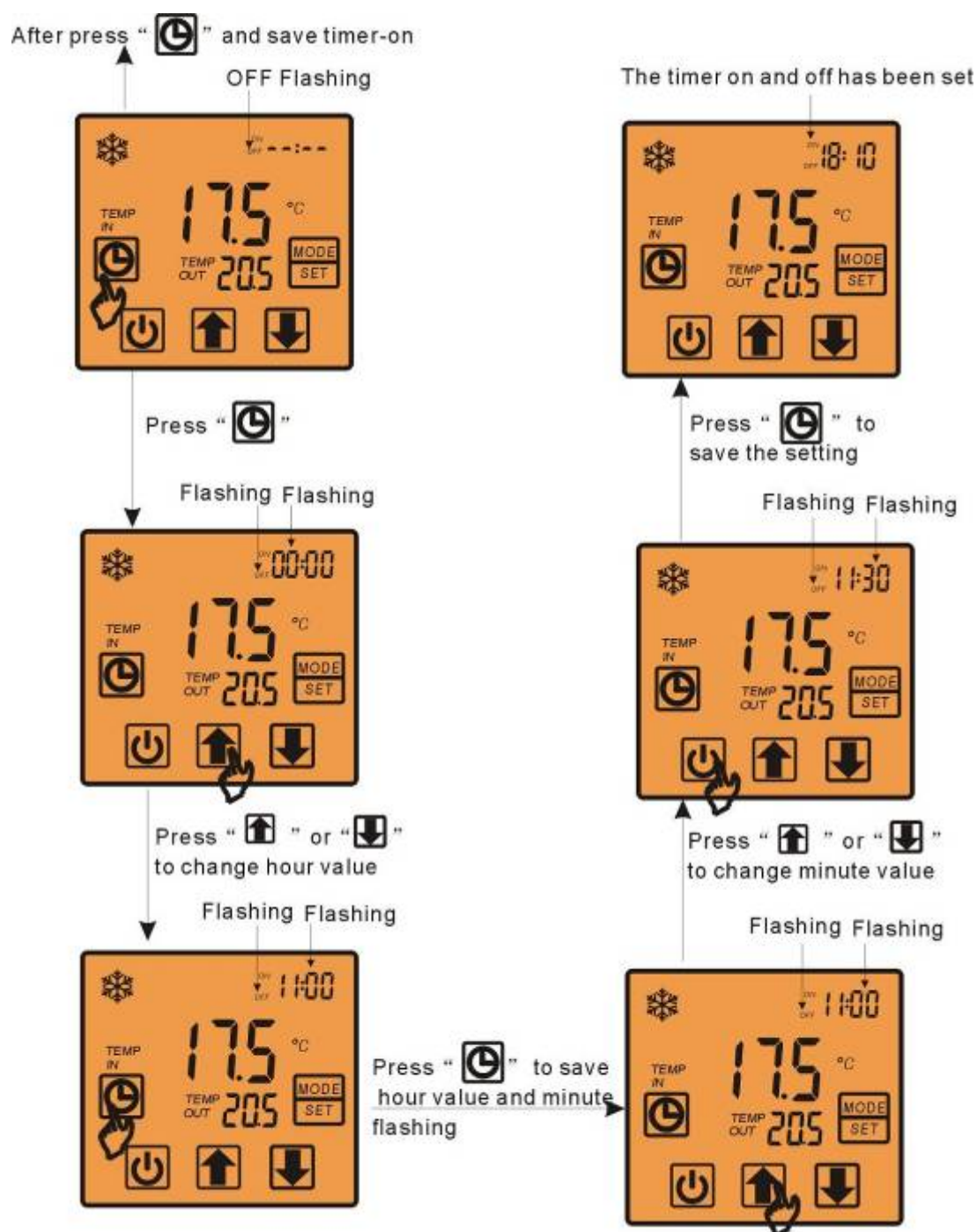
- Attention: a) After the final step, "OFF" will flash, you can press "clock" to go on to set the timer off.
- b) If you only want to set the timer on, then after the final step just press "on/off" to back to main interface.
- c) If there is no operation for 5s, system will remember clock setting and back to the main interface.

Timer off setting

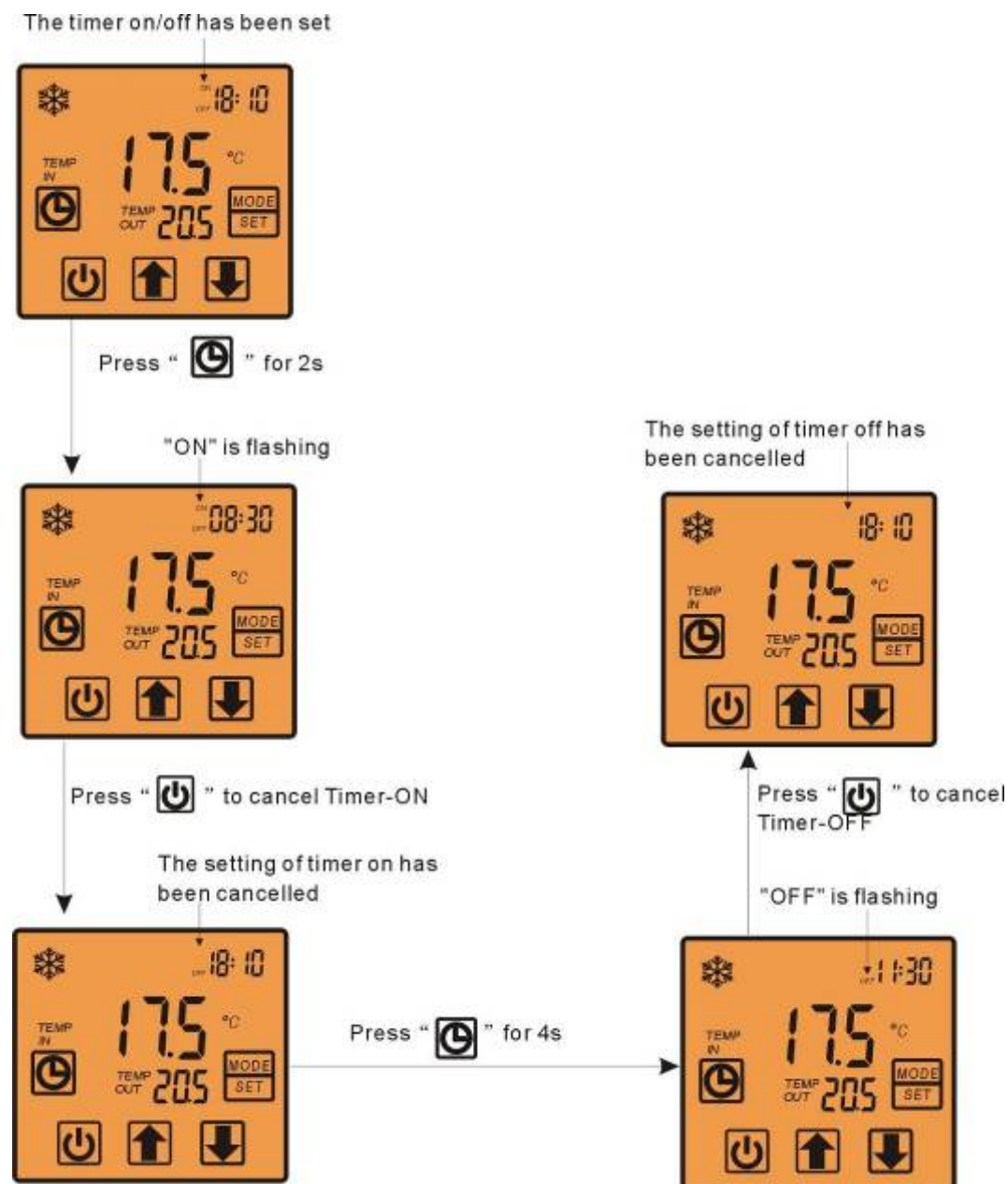
Notes:

- A) At the main interface, press “clock” till the "off" flashing, you can set the timer off.
- B) After the last step of timer-on, press “clock” can get into timer off setting interface.
- C) If there is no operation for 5s, system will remember clock setting and back to the main interface.

Now we take B) situation for example, the timer off operation as follow:



The operation of canceling timer on /off



Attention: If there is only timer-off for system, press "clock" for 4s and "OFF" flashing, then press "on/off" key to cancel timer-off.

The download operation

1) Press mode for 10s, and input password 66/22, pressing “mode” to enter parameter setting interface, as following:

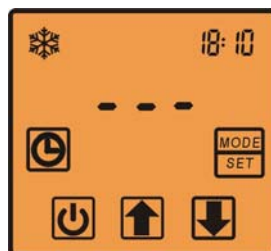


2) Upload: Press “UP” for 2s; Download: Tap “DOWN”.

During upload/ download process, the dash will flashing. When it has been done, system enter next interface.

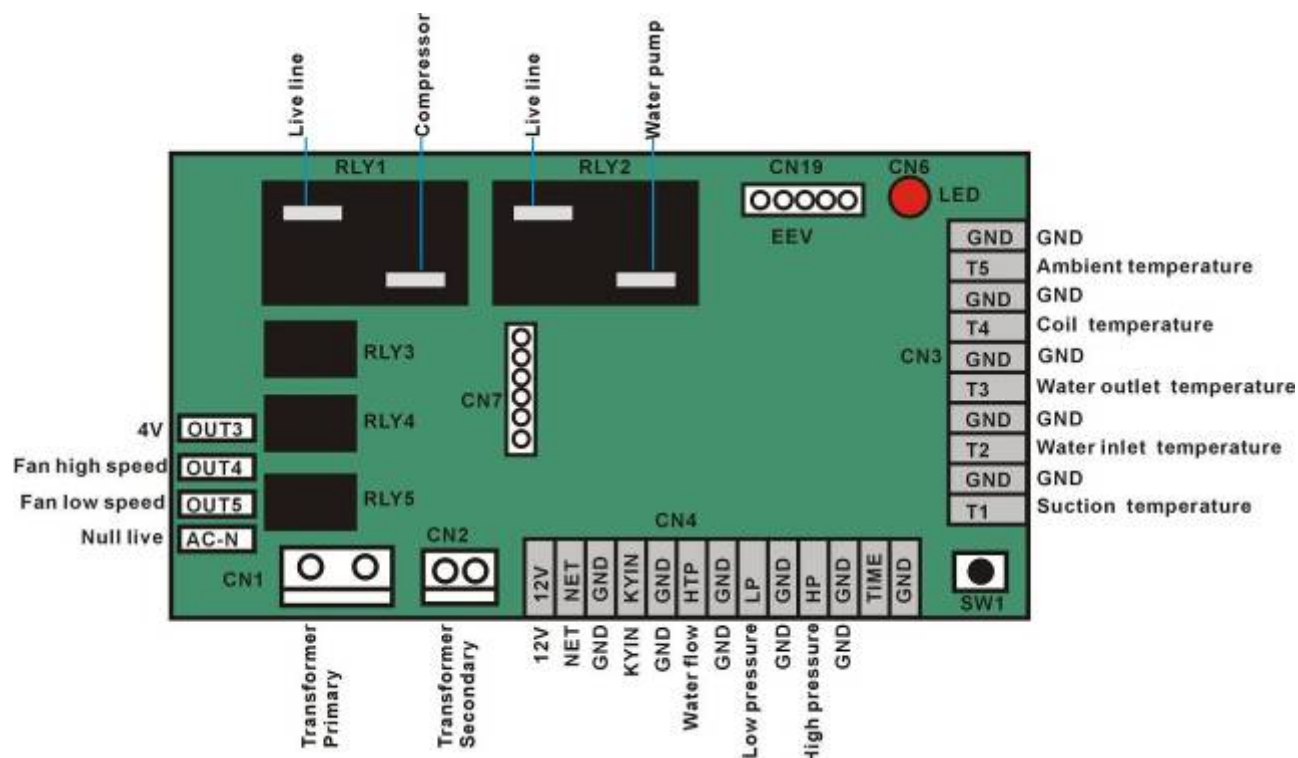


3) Waiting for a few seconds, then the screen will show the following:



4) Finally, press “on/off” to back to the main interface.

18. PCB I/O Ports description



19. Troubleshooting

There will be malfunction code showing on the controller screen when relative malfunction occurs.

Code	Failure	Reason	Solution
P01	Water inlet temp. sensor failure	Temp. Sensor is broken	Check or change it
P02	Water outlet temp. sensor failure	Temp. Sensor is broken	Check or change it
P04	Ambient temp. sensor failure	Temp. Sensor is broken	Check or change it
P05	Coil temp. sensor failure	Temp. Sensor is broken	Check or change it
P07	Suction temp. sensor 1 failure	Temp. Sensor is broken	Check or change it
E01	High pressure protection (HP)	HP switch is broken	Check or change it
E02	Low pressure protection (LP)	LP switch is broken	Check or change it
E03	Water flow protection	Flow switch is broken	Check or change it
E06	Temp. Difference between inlet and outlet	Temp. Difference > 13°C	Temp. Difference < 13°C, power off
E07	Antifreezing protection	Antifreezing temp. < 2°C	Check flow and water system
E19	Primary Antifreezing protection	Ambient temp. is too low	/
E29	Secondary Antifreezing protection	Ambient temp. is too low	/
E08	Communication failure	Communication failure between wire controller and main board	Check the connection between wire controller and main board

20. Parameter

◆ Parameter table

NO.	Type	parameter and description	Setting value		Default level
			Code	Setting	
1	d	Start defrosting temperature	d01	-30-0℃	F/U
2		End defrost temperature	d02	0-30℃	F/U
3		Defrosting cycle	d03	1-90min	F/U
4		Maximum defrosting time	d04	1-20min	F/U
5	E	EEV mode	E01	0-1	F/U
6		Super heat	E02	-20-20℃	F
7		Initial place	E03	0-500	F
8		Minimum place	E04	0-500	F
9		Defrost place	E05	0-500	F
10		Cooling place	E06	0-500	F
11	F	Fan parameter	F01	0-4	F/U
12		Coil temp. in high speed fan mode(Cooling)	F02	-15-60℃	F
13		Coil temp. in low speed fan mode(Cooling)	F03	-15-60℃	F
14		Coil temp. when the fan stop (Cooling)	F04	-15-60℃	F
15		Coil temp. in high speed fan mode(Heating)	F05	-15-60℃	F
16		Coil temp. in low speed fan mode(Heating)	F06	-15-60℃	F
17		Coil temp. when the fan stop(Heating)	F07	-15-60℃	F
18		Fan start low speed running time	F08	0-23h	F
19		Fan stop low speed running time	F09	0-23h	F
20		Fan speed control temp.	F10	0-1	F
21	h	Automatic restarting	H01	0-1	F
22		Model(cooling only/AUTO/heating only)	H02	0-2	F/U
23		Temperature unit transformation	H03	0-1	F
24	P	Water pump model	P01	0-2	F/U
25		Water pump running cycle	P02	0-120min	F/U
26		Water pump running time	P03	0-30min	F/U
27		Delay in switching on the compressor after switching on the pump	P04	0-30min	F/U
28	r	Inlet water setting temp. (cooling)	r01	r08-r09	F
29		Inlet water setting temp. (Heating)	r02	r10-r11	F
30		Target setting temp. (Auto mode)	r03	r08-r11	F
31		Cooling differential	r04	0-10℃	F
32		Cooling stop differential	r05	0-10℃	F
33		Heating differential	r06	0-10℃	F

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34	r	Heating stop differential	r07	0-10°C	F
35		Minimum set point in Cooling	r08	-30-r09°C	F
36		Maximum Cooling set point	r09	r08-80°C	F
37		Minimum Heating set point	r10	-30-r11°C	F
38		Maximum Heating set point	r11	r11-80°C	F
39	S	On/off switch	S01	CL/OP	F/U
40		Water Flow switch	S02	CL/OP	F/U
41		System LP	S03	CL/OP	F/U
42		System HP	S04	CL/OP	F/U
43		Mode switch	S05	CL/OP	F/U
44	t	Suction temp.	T01	-30~99°C	F/U
45		Inlet water temp.	T02	-30~99°C	F/U
46		Outlet water temp.	T03	-30~99°C	F/U
47		Coil temp.	T04	-30~99°C	F/U
48		Ambient temp.	T05	-30~99°C	F/U
49	O	Compressor output	O1	CL/OP	F/U
50		Circulate pump output	O2	CL/OP	F/U
51		4-way valve output	O3	CL/OP	F/U
52		Fan output (High speed)	O4	CL/OP	F/U
53		Fan output (Low speed)	O5	CL/OP	F/U
54		Electronic expansion valve output	O6	0~500	F/U

◆ Description of the parameters

D——Defrost parameter

D01——Start defrost temperature

To start the defrost cycle; the condition must be valid for the time d03.

D02——End defrost temperature

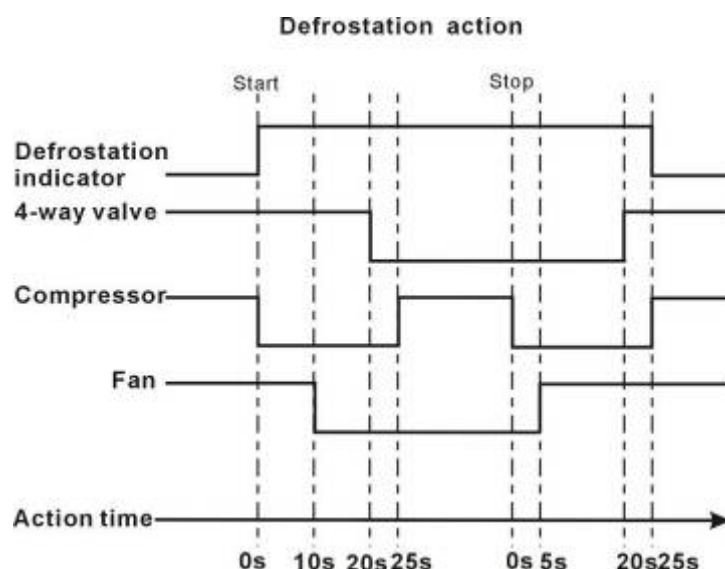
Establishes the temperature above which the defrost cycle ends.

D03——Defrosting cycle

Represents delay between two successive defrost cycle. The first time, when coil temperature is lower than D01, there must be valid for the time d03 to start defrost.

D04——Max.defrosting duration

Represents the maximum duration of the defrost cycle (the defrost ends when the maximum duration has been arrived, even if the defrost hasn't finished)



Attention: The situation of defrostation abnormal end

- 1) Shut off the unit during defrosting, system will continue running defrost until it has finished.
- 2) HP switch has broken during defrosting, then unit will be shut off and show HP malfunction. After recovering it, system enters to normal heating mode.
- 3) LP switch has broken during defrosting, the unit will skip LP malfunction and exit defrosting and back to normal heating mode, then system will check LP switch after 5min.
- 4) Flow switch has broken during defrosting, then unit will be shut off and show Flow Malfunction. After recovering this malfunction, system goes on defrosting.
- 5) Exhaust temperature is too high during defrosting, then unit will be shut off and show this malfunction. After recovering it, system goes on defrosting.
- 6) Temperature difference between inlet and outlet during defrosting, then unit will be shut off and show this malfunction. After recovering it, system goes on defrosting.
- 7) System show Antifreezing protection during defrosting, then unit will be shut off and show this malfunction. After recovering it, system goes on defrosting.

E—EEV parameter

E01—EEV mode

E01=0: EEV is running by manual operation;

E01=1: EEV is running by automatic operation;

E02—Target Super heat

E03—Initial position

If E01=0, represents expansive valve fix this position always.

If E01=1, represents expansive valve initiation position

E04—Minimum position

E05—Defrost position

Fix the EEV position during system is defrosting.

E06—Cooling position

Fix the EEV position during system at cooling mode.

F——Fan parameter

Normally, Fan will start up 5s ahead of Compressor and 30s later to shut off. When at defrosting, Fan running situation is according to defrosting control.

F01——Fan parameter

F01=0: in low speed fan mode;

F01=1: in high speed fan mode;

F01=2: the fan running modes depend on coil or ambient temperature (F02-F07);

Attention: The temperature probe is decided by F10.

F01=3: the fan runs at low speed depends on time (F08-F09), the fan runs at high speed during the other time;

F01=4: the fan running speed depends on F02 and F03.

F02——Coil or ambient temperature set point for high speed fan mode (Cooling)

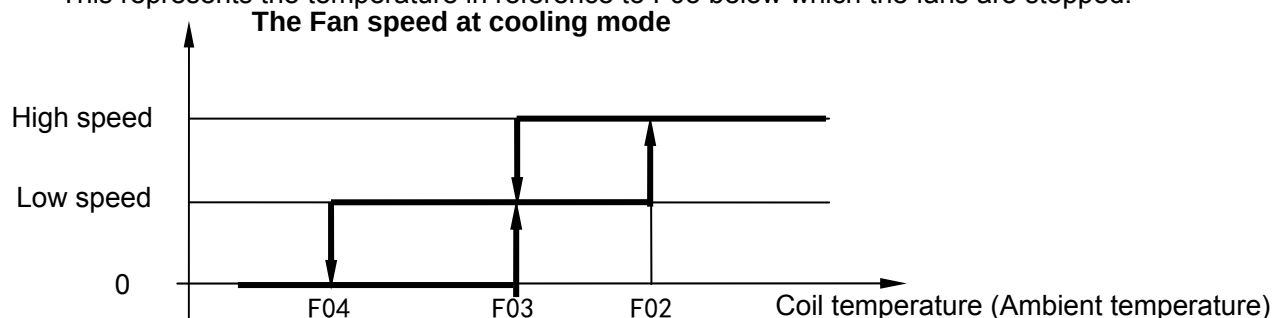
This represents if the temperature above F02, the fan will on high speed (Cooling)

F03——Coil or ambient temperature set point for low speed fan mode (Cooling)

This represents if the temperature below which the fans remain on at low speed (Cooling)

F04——Coil or ambient temperature set point for the fan stop (Cooling)

This represents the temperature in reference to F03 below which the fans are stopped.



F05——Coil or ambient temperature set point for high speed fan mode (Heating)

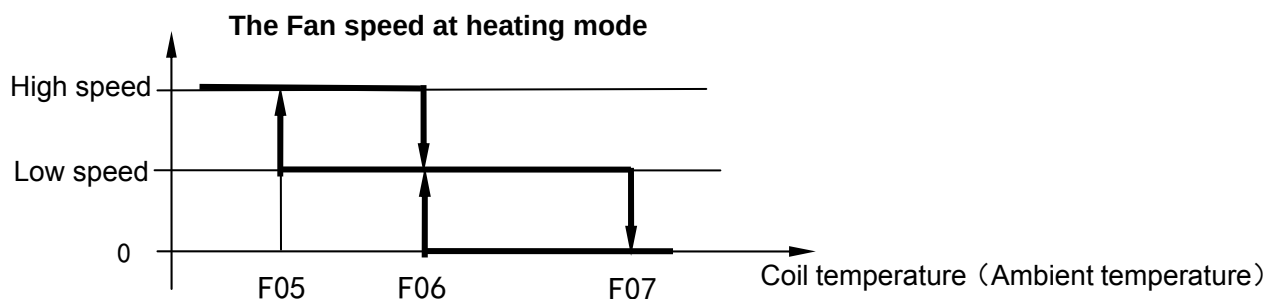
This represents the temperature above which the fans remain on at high speed (Heating)

F06——Coil or ambient temperature set point for low speed fan mode (Heating)

This represents the temperature below which the fans remain on at low speed (Heating)

F07——Coil or ambient temperature set point for the fan stop (Heating)

This represents the temperature in reference to F06 below which the fans are stopped.



F08——Fan start low speed running time (Just for F01=3)

F09——Fan stop low speed running time (Just for F01=3)

F10——Fan speed control temp.

When F10=0, Fan speed is decided by coil temperature;

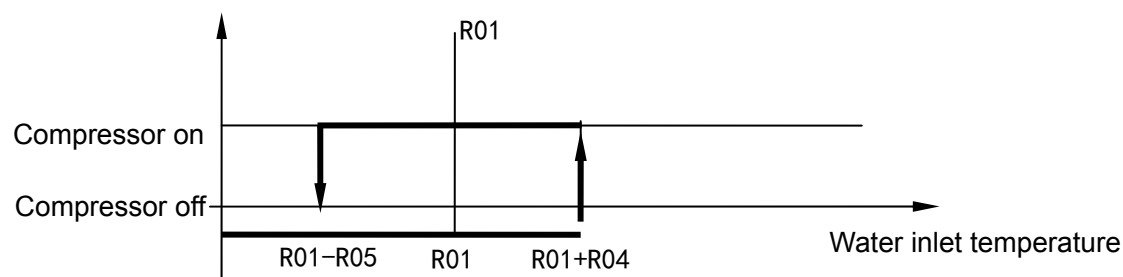
When F10=1, Fan speed is decided by ambient temperature.

Swimming pool heat pump

R05——Stop differential of cooling

This represents the difference between R01 and stop cooling point.

Compressor action at cooling mode



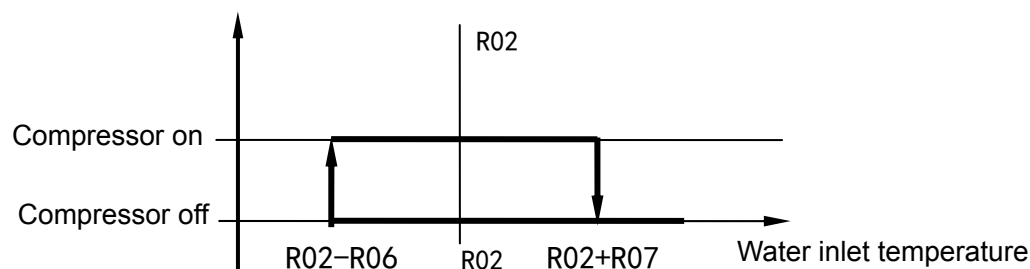
R06——Start differential of heating

This represents the difference between R02 and start heating point.

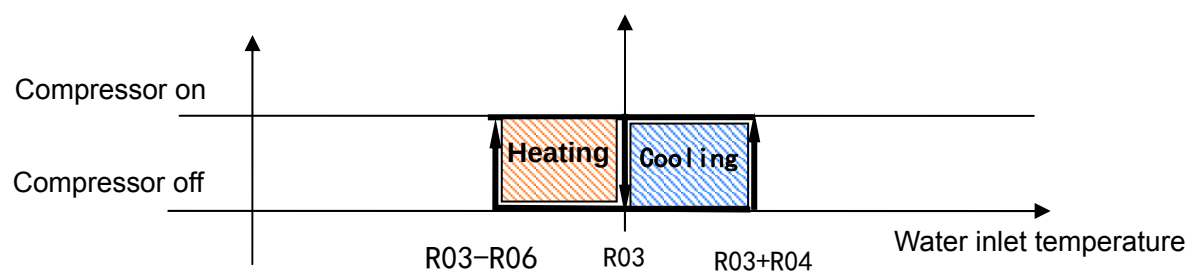
R07——Stop differential of heating

This represents the difference between R02 and stop heating point.

Compressor action at heating mode



Compressor action at Automatic mode



R08——Min. set point in Cooling

Establishes the minimum limit for setting the Cooling set point

R09——Max. Cooling set point

Establishes the maximum limit for setting the Cooling set point

R10——Min. Heating set point

Establishes the minimum limit for setting the Heating set point

R11——Max. Heating set point

Establishes the maximum limit for setting the Heating set point

Appendix I

NTC R-T Table (R25=5K Ω B25/50=3470K)

T(°C)	R(K Ω)		T(°C)	R(K Ω)		T(°C)	R(K Ω)
-30.0	63.7306		14.0	7.7643		58.0	1.5636
-29.0	60.3223		15.0	7.4506		59.0	1.5142
-28.0	57.1180		16.0	7.1513		60.0	1.4666
-27.0	54.1043		17.0	6.8658		61.0	1.4206
-26.0	51.2686		18.0	6.5934		62.0	1.3763
-25.0	48.5994		19.0	6.3333		63.0	1.3336
-24.0	46.0860		20.0	6.0850		64.0	1.2923
-23.0	43.7182		21.0	5.8479		65.0	1.2526
-22.0	41.4868		22.0	5.6213		66.0	1.2142
-21.0	39.3832		23.0	5.4048		67.0	1.1771
-20.0	37.3992		24.0	5.1978		68.0	1.1413
-19.0	35.5274		25.0	5.0000		69.0	1.1068
-18.0	33.7607		26.0	4.8108		70.0	1.0734
-17.0	32.0927		27.0	4.6298		71.0	1.0412
-16.0	30.5172		28.0	4.4566		72.0	1.0100
-15.0	29.0286		29.0	4.2909		73.0	0.9800
-14.0	27.6216		30.0	4.1323		74.0	0.9509
-13.0	26.2913		31.0	3.9804		75.0	0.9228
-12.0	25.0330		32.0	3.8349		76.0	0.8957
-11.0	23.8424		33.0	3.6955		77.0	0.8695
-10.0	22.7155		34.0	3.5620		78.0	0.8441
-9.0	21.6486		35.0	3.4340		79.0	0.8196
-8.0	20.6380		36.0	3.3113		80.0	0.7959
-7.0	19.6806		37.0	3.1937		81.0	0.7730
-6.0	18.7732		38.0	3.0809		82.0	0.7508
-5.0	17.9129		39.0	2.9727		83.0	0.7293
-4.0	17.0970		40.0	2.8688		84.0	0.7086
-3.0	16.3230		41.0	2.7692		85.0	0.6885
-2.0	15.5886		42.0	2.6735		86.0	0.6690
-1.0	14.8913		43.0	2.5816		87.0	0.6502
0.0	14.2293		44.0	2.4934		88.0	0.6320
1.0	13.6017		45.0	2.4087		89.0	0.6144
2.0	13.0057		46.0	2.3273		90.0	0.5973
3.0	12.4393		47.0	2.2491		91.0	0.5808
4.0	11.9011		48.0	2.1739		92.0	0.5647
5.0	11.3894		49.0	2.1016		93.0	0.5492
6.0	10.9028		50.0	2.0321		94.0	0.5342

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7.0	10.4399		51.0	1.9656		95.0	0.5196
8.0	9.9995		52.0	1.9015		96.0	0.5055
9.0	9.5802		53.0	1.8399		97.0	0.4919
10.0	9.1810		54.0	1.7804		98.0	0.4786
11.0	8.8008		55.0	1.7232		99.0	0.4658
12.0	8.4385		56.0	1.6680		100.0	0.4533
13.0	8.0934		57.0	1.6149			

- 1) When there is some malfunction, test resistance value by multimeter, and compare the practical temperature with the above table, then you will know whether this NCT resistance is OK or not.
- 2) Generally, from above table, you can know the temperature by testing NTC resistance value.

PHNIX

Guang Dong PHNIX eco-energy solution Ltd.